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Online Matching and Contention Resolution for Edge Arrivals with Vanishing Probabilities

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Abstract

We study the performance of sequential contention resolution and matching algorithms on random graphs with vanishing edge probabilities. When the edges of the graph are processed in an adversarially-chosen order, we derive a new OCRS that is 0.382-selectable, attaining the "independence benchmark" from the literature under the vanishing edge probabilities assumption. Complementary to this positive result, we show that no OCRS can be more than 0.390-selectable, significantly improving upon the upper bound of 0.428 from the literature. We also derive negative results that are specialized to bipartite graphs or subfamilies of OCRSs. Meanwhile, when the edges of the graph are processed in a uniformly random order, we show that the simple greedy contention resolution scheme which accepts all active and feasible edges is 1/2-selectable. This result is tight due to a known upper bound. We then show that when the algorithm can choose the processing order, a slight tweak to the random order—give each vertex a random priority and process edges in lexicographic order—results in a strictly better contention resolution scheme that is $1 - \ln(2 - 1/e) \approx 0.510$-selectable. Moreover, we show that this bound is tight over any sequential contention resolution scheme, even one which may adaptively choose the order in which it processes edges. This provides a separation from the 0.544 upper bound for offline contention resolution implied by the classic result of Karp and Sipser. Our positive results also apply to online matching on 1-uniform random graphs with vanishing (non-identical) edge probabilities, extending and unifying some results from the random graphs literature.

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1 Introduction

A common problem faced by two-sided matching platforms such as ride-sharing apps and online labor marketplaces is to decide how to assign supply to demand arriving in an online fashion, while accounting for the value of the assignment and uncertainty about future demand (Özkan and Ward, 2020; Hu and Zhou, 2022; Feng et al., 2024; Pollner et al., 2022). Since assigning a worker to one job prevents that worker from taking another job at the same time, platforms must compare the value of a match available now with the value of other matches that it might make later.

Such decisions can be modeled as *Bayesian online matching problems*, where we have some distributional knowledge of the value of future matches, but we learn their actual realized values in an online fashion. A common strategy for these problems is to first solve a *fluid (ex-ante) relaxation*, which makes fractional decisions maximizing the expected value subject to the feasibility constraints holding only in expectation (this is typically easy, since the relaxation is using a linear or convex program), and then find a way to convert its solution into an online policy (Alaei et al., 2012; Alaei, 2014; Jiang et al., 2025; Balseiro et al., 2024).

Generally speaking, it is impossible for an online policy to match the value of the fluid relaxation due to the stochasticity of the realized arrivals. The goal is therefore to obtain some constant fraction of this value while satisfying the feasibility constraints in every realization. *Contention resolution schemes* are a technical tool for achieving this goal (Feldman et al., 2021).

Motivated by these concerns, we study variants of contention resolution schemes for matchings on graphs. Formally, we have a graph $G = (V, E)$, and each edge $e \in E$ has a random state $X_e \in \{0, 1\}$ that is unknown a priori, but known to be *active* (i.e., $X_e = 1$) independently with probability x_e. This x_e represents the probability with which a fluid relaxation of an allocation problem would accept the edge.

The edges arrive one-by-one, at which point their random state is revealed. If the edge is not active, then it is discarded; if the edge is active, then an immediate decision must be made about whether to select the edge. The edge is only feasible to select if it does not share a vertex with any previously-selected edge. That is, the set of edges selected must form a *matching* in the graph.

The probabilities $(x_e)_{e \in E}$ form a fractional matching in the graph. That is,

$$\sum_{e \in \partial(v)} x_e \leq 1 \qquad \forall v \in V, \tag{1.1}$$

which says that the expected number of active edges incident to any vertex v is at most 1. The selection problem can be interpreted as "rounding" a fractional matching into an integer one, where the rounding occurs online, and only active edges can be rounded up. The goal is to provide a uniform guarantee that every edge $e \in E$ is selected with probability at least cx_e, for a constant c as large as possible. An algorithm that provides such a guarantee is called a c-selectable contention resolution scheme.

To make the connection between contention resolution schemes and applications in Bayesian online matching problems precise, consider a model with a fixed set of jobs and workers, where each worker can complete at most one job. Vertices represent jobs and workers, and an edge $e = (u, v)$ represents assigning job u to worker v. We have distributional knowledge of W_e, the non-negative value obtained by making this assignment, and assume that the W_e are independent of each other. We learn the actual realized value of each pair in an online fashion and must immediately decide whether that pair should belong to the overall matching, while attempting to maximize the total value of the matching constructed at the end of the process. This is the prophet inequality matching problem (Feldman et al., 2021; Gravin and Wang, 2023; Ezra et al., 2022).

For simplicity, we present the ex-ante relaxation for the setting when each W_e is a weighed Bernoulli random variable; that is, $W_e \in \{0, w_e\}$, where $\Pr[W_e = w_e] = p_e$ for some fixed value $w_e \geq 0$ and probability $p_e \in [0, 1]$. An ex-ante relaxation introduces a variable $y_e \in [0, 1]$ for each edge e, which should be thought of as the ideal probability of selecting edge e, given $W_e = w_e$. By also imposing expected matching constraints on the vertices, this leads to the following linear program:

$$\begin{aligned} \max \quad & \sum_{e \in E} w_e p_e y_e \\ \text{s.t.} \quad & \sum_{e \in \partial(v)} p_e y_e \leq 1 \quad \forall v \in V, \\ & 0 \leq y_e \leq 1 \qquad \forall e \in E. \end{aligned} \tag{1.2}$$

The relaxation upper bounds the expected value obtainable by any matching algorithm, including the "prophet benchmark" which is revealed the instantiations of $(W_e)_{e \in E}$ upfront. For an optimal solution to this program, say $(y_e)_{e \in E}$, notice that if $x_e = y_e x_e$, then the constraints on $(x_e)_{e \in E}$ are exactly (1.1).

We now explain how to use a contention resolution scheme for $(x_e)_{e \in E}$ to design a policy for the prophet inequality matching problem. When edge e arrives, drawn a Bernoulli Y_e of mean y_e independently, and declare edge e *active* if $Y_e = 1$ and $W_e = w_e$. Notice that each edge is active independently with overall probability x_e, and so we can apply a contention resolution scheme to decide whether to select e. As every edge e is selected with probability at least c conditional on being active, we get

$$\mathbb{E}\left[\sum_{e \in E} w_e \cdot \mathbf{1}_{e \in \mathcal{M}}\right] \geq c \sum_{e \in E} w_e \cdot x_e = c \sum_{e \in E} w_e \cdot p_e \cdot y_e,$$

where $\mathcal{M}$ is the matching selected by the scheme. Thus, we have found an online algorithm that obtains a c fraction of the ex-ante relaxation's value. While we have presented the reduction for the special case of weighted Bernoulli's, an almost identical argument extends to the case of arbitrary random variables by replacing (1.2) with a convex program (see (Feldman et al., 2021; Lee and Singla, 2018) for details).

Similar reductions exist for sequential posted-price mechanisms (Chawla et al., 2010; Yan, 2011; Adamczyk and Włodarczyk, 2018; Pollner et al., 2022; Derakhshan et al., 2026) and stochastic probing (Gupta and Nagarajan, 2013; Gupta et al., 2016, 2017; Borodin and MacRury, 2025). In stochastic probing, an edge's state is revealed only after it is probed, and a probed active edge must be selected. For matching applications, probing can represent offering a job to a worker, with additional *patience constraints* limiting the number of offers a work is willing to consider, as originally introduced by Chen et al. (2009); Bansal et al. (2012). These offers can also depend on some monetary payment Pollner et al. (2022); Derakhshan et al. (2026), in which case the acceptance probability of the worker taking the job may depend on the payment offered. We defer to relevant works for further details on these reductions.

In applications of contention resolution, the order in which edges are processed is important. This leads to the following problem variants.

- **Adversarial Order**: the edges are processed in a worst-case[1] order. An online selection algorithm in this setting is called an Online Contention Resolution Scheme (OCRS).

[1]We assume that this order is chosen by an adversary who knows the description of the algorithm. They cannot however change the order after the algorithm begins its execution.

- **Random Order**: the edges are processed in a uniformly random order. An online selection algorithm in this setting is called a Random-order Contention Resolution Scheme (RCRS).
- **Free Order**: the edges are processed in an order chosen by the algorithm. An online selection algorithm in this setting is called a Free-order Contention Resolution Scheme (FO-CRS).

We note that contention resolution was originally studied in the offline setting (Chekuri et al., 2014), where all random states are revealed before making any selections.

In this paper we consider OCRS, RCRS, and FO-CRS in the *vanishing regime*, which was previously resolved by Nuti and Vondrák (2024) in the offline setting. The idea of this regime is to assume every edge is active with a probability less than some small ε, and provide limiting guarantees as $\varepsilon \to 0$. This is an important regime for the following reasons.

- **Precedent**: x_e being small has also been a special case of interest in other models of online matching:
  - Literature on the online matching with stochastic rewards model (Mehta and Panigrahi, 2012; Mehta et al., 2014; Huang and Zhang, 2020; Goyal and Udwani, 2023; Huang et al., 2023) generally focuses on this regime.
  - Motivated by online edge-coloring, rounding fractional matchings online with small x_e has also been studied extensively. In this problem, there is no active/inactive status associated with edges. Instead, the fractions x_e are revealed in an online fashion, and when x_e is revealed, we must immediately decide whether e should be included in the matching. The goal is to select a matching which contains edge e with probability at least cx_e. Cohen et al. (2019) showed that for bipartite one-sided vertex arrivals (when a vertex arrives, the values of all its incident edges are revealed), a $1 - o(1)$ guarantee is attainable, where $o(1)$ is a function that tends to 0 as $\max_{e \in E} x_e \to 0$. For edge arrivals, Kulkarni et al. (2022) proved a rounding guarantee of $1 - 1/e - o(1)$, and this was recently improved to $1 - o(1)$ by Blikstad et al. (2024).
  - The random graphs literature analyzes what happens when $|V| \to \infty$, with $x_e = p \to 0$ for all e if the graph is sparse, which is a special case of the vanishing regime with equal probabilities. We relate our results to this literature at the end of Subsection 1.1.
- **Tightness**: The vanishing or "Poisson" regime is often where worst-case examples occur in online Bayesian selection, especially in problems like contention resolution where one is competing against an ex-ante benchmark. The simplest example arises in RCRS for rank-1 matroids (Lee and Singla, 2018), where the guarantee c cannot be larger than $1 - 1/e$ due to the vanishing regime. More specifically for matchings in graphs, three settings where optimal contention resolution schemes are known to have tight examples that lie in the vanishing regime: offline monotone bipartite contention resolution (Bruggmann and Zenklusen, 2022), OCRS for two-sided vertex arrivals (Ezra et al., 2022), and, RCRS for two-sided bipartite vertex arrivals (MacRury and Ma, 2024).

  Our results suggest that a plausible strategy for optimal contention resolution is to start with an algorithm for the vanishing regime, and then add adjustments as necessary to to obtain the optimal algorithm for the non-vanishing regime, even though we acknowledge the second step may not be easy. Bruggmann and Zenklusen (2022, Lemma 7, Appendix A.1) already show how to reduce to the vanishing regime if the reduced graph can have parallel edges by splitting each edge e into k parallel copies with values x_e/k; however, our results are for graphs *without parallel edges*.

We are therefore interested in the following question:

What is the optimal OCRS, RCRS, and FO-CRS in the vanishing regime?

1.1 Outline of Results

We say that an OCRS (resp. RCRS, FO-CRS) is *c-selectable* if it selects every edge $e \in E$ with probability at least cx_e, under an adversarial (resp. random, free) arrival order, for all graphs $G = (V, E)$ and vectors $(x_e)_{e \in E}$ satisfying (1.1). We say that it is *c-selectable in the vanishing regime* if it is $(c - f(\max_{e \in E} x_e))$-selectable for all G and $(x_e)_{e \in E}$ satisfying (1.1), where f is a function satisfying $\lim_{\varepsilon \to 0} f(\varepsilon) = 0$. We call constant c the *selection guarantee* or just guarantee, which lies in [0,1]. We use *selectability* to refer to largest c for which an OCRS, RCRS, or FO-CRS can be c-selectable, possibly in the vanishing regime. By *optimal* contention resolution scheme, we refer to one that achieves the selectability for a particular setting (it does not have to be instance-optimal).

Theorem 1.1. *There is a polytime OCRS that is $\frac{3-\sqrt{5}}{2} \approx 0.382$-selectable in the vanishing regime.*

We let α denote the constant $\frac{3-\sqrt{5}}{2}$, which is the smaller real number satisfying $\alpha = (1-\alpha)^2$. This is the guarantee that would be achieved if we already knew for a fact that the events that the endpoints of an arriving edge are unmatched were independent (Ezra et al., 2022, §4.2). We are able to show that this guarantee is in fact achieved in the vanishing regime. Formalizing the way in which small activeness probabilities x_e imply approximate independence is the main technical challenge in this work, as we explain in Section 2. We also note that we are using a different OCRS for selecting edges than Ezra et al. (2022), because their algorithm has a sophisticated behavior whose independence properties are difficult to analyze.

Theorem 1.2.

1. *No OCRS can be better than $\gamma \approx 0.393$-selectable, even in the vanishing regime over tree graphs.*
2. *No OCRS can be better than $\beta \approx 0.390$-selectable, even in the vanishing regime.*
3. *No OCRS that satisfies a concentration property (see Subsection 4.2 and Section E) can be better than $\alpha = \frac{3-\sqrt{5}}{2}$-selectable, even in the vanishing regime over tree graphs.*

In parts (1) and (2) of Theorem 1.2, we establish upper bounds of 0.393 and 0.390 over vanishing tree graphs (which are bipartite graphs) and vanishing general graphs, respectively. Part (3) of Theorem 1.2 says that the 'independence benchmark' of α is tight over OCRSs that satisfy a certain concentration property. This property holds for all OCRSs in the literature (we discuss this in greater detail in Section E). Moreover, if our objective was instead to maximize expected size of the algorithm's matching, then a standard Doob martingale argument implies that the optimal policy is concentrated. Due to these points, we believe it is intuitive to posit that an optimal OCRS should satisfy this concentration property, even though we are not able to prove it. As we explain in Section 2, these upper bounds use different arguments than existing ones in the literature, and also offer a significant improvement. Indeed, the previous upper bounds were $4/9 \approx 0.444$ (Gravin and Wang, 2019) followed by $3/7 \approx 0.428$ (Correa et al., 2022) for bipartite graphs, and 0.4 (MacRury et al., 2025) for general graphs (this final result did not use vanishing edge values).

Turning to RCRS and FO-CRS, we will assume that G is *1-regular*, which makes it easier to describe optimal algorithms. This is the assumption that $\sum_{e \in \partial(v)} x_e = 1$ for all v, i.e. (1.1) is tight, which has been shown to be the worst case via reductions (see Section 5). In the random order setting, we show that the greedy [2]CRS, which accepts every arriving edge that is active and feasible is 1/2-selectable:

Theorem 1.3. *The greedy CRS is a $\frac{1}{2}$-selectable RCRS in the vanishing regime.*

Theorem 1.3 is tight because no RCRS can be better than 1/2-selectable, even in the vanishing regime (MacRury et al., 2025). It is worth noting that the 0.474-selectable RCRS of MacRury et al. (2025), which is state-of-the-art for non-vanishing edges, behaves identically to the greedy algorithm in the vanishing regime, and hence is 1/2-selectable. This contrasts with the OCRS variant, where the state-of-the-art algorithm originating from Ezra et al. (2022) cannot attain the selection guarantee of α from the vanishing regime (its guarantee is upper-bounded by 0.361, as shown by MacRury et al. (2025)).

Theorem 1.4. *There exist random orderings of edges under which the greedy CRS is a $1 - \ln\left(2 - \frac{1}{e}\right) \approx 0.510$-selectable FO-CRS in the vanishing regime.*

Theorem 1.4 establishes a strict separation between free and random order, for edge arrivals in graphs in the vanishing regime. To our knowledge, results that show how to improve upon random order are generally rare. In stochastic probing and price of information problems on graphs in which the algorithm can choose the order, many results (Gamlath et al., 2019; Fu et al., 2021) still defer to analyzing the random order setting at some point.

Theorem 1.5. *No FO-CRS can be better than better than $1 - \ln\left(2 - \frac{1}{e}\right)$ selectable in the vanishing regime.*

We prove Theorem 1.5 by showing that on the complete bipartite graph with $2n$ vertices and uniform edge values, no FO-CRS can construct a matching whose expected size is asymptotically greater than $(1 - \ln\left(2 - \frac{1}{e}\right))n$ as $n \to \infty$. A classic result of Karp and Sipser (1981) showed that the optimal matching of this same input has expected size asymptotically $\approx 0.544n$, and Nuti and Vondrák (2024) showed that there is an offline CRS with a selection guarantee of ≈ 0.544 in the vanishing regime. Thus, Theorem 1.5 gives a strict separation between free-order contention resolution and offline contention resolution in the vanishing regime.

Relation to oblivious matching and query-commit. In the edge-weighted *oblivious matching problem*, the edges have known weights, however whether they exist or not is determined by an unknown (arbitrary) distribution. The algorithm probes edges in an order of its choosing, subject to the constraint that if an edge is found to exist, it must be added to the matching. By randomly perturbing the edge weights and executing greedily, Tang et al. (2023) prove a 0.501-approximation ratio against the matching of largest expected weight. The oblivious matching setting generalizes the edge-weighted *query-commit matching problem*, the latter of which has been studied extensively, beginning with the work of Chen et al. (2009); Bansal et al. (2012) (see Borodin and MacRury (2025) for a detailed overview of this problem and its variants). In this problem, the edges are realized independently according to known probabilities. Fu et al. (2021) design a 0.533-selectable random-order *vertex arrival* OCRS which they use with the linear program of Costello et al. (2012); Gamlath et al. (2019) to get an approximation ratio of 0.533 for this problem. (This selection guarantee, and thus approximation ratio, was recently improved to 0.535 by MacRury and Ma (2024)).

[2]This is not to be confused with the notion of a greedy OCRS from Feldman et al. (2021).

Assuming the edge probabilities are vanishing, we can apply our free-order OCRS to get a 0.510 approximation ratio for this problem, and thus beat 1/2 in a different way.

We summarize our results and how they position in the literature in Table 1.

Selectability Bounds	General Edge Values x_e	Vanishing Edge Values $x_e \to 0$
OCRS	≥ 0.344 [3] ≤ 0.4 [3] $\to \leq$ **0.390** [§4]	$\to \geq$ **0.382** [§3] ≤ 0.428 [2] $\to \leq$ **0.390**, ≤ 0.428 [2] $\to \leq$ **0.393** (bipartite) [§4]
RCRS	≥ 0.474 [3] ≤ 0.5 [3]	$\to \geq$ **0.5** [§5] ≤ 0.5 [3]
FO-CRS	≥ 0.474 [3] ≤ 0.544 [1] $\to \leq \approx$ **0.510** [§6]	$\to \geq \approx$ **0.510** [§5] ≤ 0.544 [1] $\to \leq \approx$ **0.510** [§6]

Table 1: Summary of known results, with new results **bolded**. "$\geq$" refers to lower bounds on c (algorithmic results), "$\leq$" refers to upper bounds (impossibility results), and arrows indicate improvement from state of the art. [1] is Karp and Sipser (1981), [2] is Correa et al. (2022), and [3] is MacRury et al. (2025).

Relation to matching in random graphs. Dyer et al. (1993) show, among other results, that the greedy algorithm leaves 50% of the vertices matched in expectation when it is executed on an Erdős–Rényi random graph with n vertices and $n/2$ edges (i.e. average degree 1), as $n \to \infty$, if it processes the edges in a uniformly random order. They show that a modified order, in which a vertex is first chosen uniformly at random and then an edge incident to it (if any) is chosen uniformly at random, leaves $1 - \ln\left(2 - \frac{1}{e}\right) \approx 51\%$ of the vertices matched. Their arguments are combinatorial. Mastin and Jaillet (2013) show that these answers (under either order) are unchanged if one considers a complete bipartite Erdős–Rényi graph instead, using the differential equation method (Wormald et al., 1999). Our Theorems 1.3 and 1.4 can be viewed as a stronger version of these results (in the special case with average degree 1), because:

- We allow for an *arbitrary* graph with vanishing probabilities and average degree 1, whereas both the results for complete and complete bipartite Erdős–Rényi graphs assume that edge probabilities are equal;
- Despite the asymmetry in our probabilities, we provide *uniform* guarantees to every edge for being selected, and show that the guarantee does not get worse. Put another way, our guarantee holds for max-weight matching instead of just max-cardinality matching.

As explained in Section 2, we also use a differential equation method based on the Galton-Watson trees from Nuti and Vondrák (2024). Interestingly, the random edge orderings we consider in Theorem 1.4 are different[3] from the modified order described above.

Further related work. In the classical revenue management formulation of a resource allocation problem (Gallego and van Ryzin, 1994), and in many models of matching problems (see, for instance Aouad and Sarıtaç (2022) or Özkan and Ward (2020)) demand arrives according to a Poisson process. Since a Poisson process of arrivals is a limit of tiny Bernoulli arrivals, results in the vanishing regime are related to problems with Poisson arrivals (see Ma et al., 2026; Aminian et al., 2026). However, a formal reduction requires results for graphs with parallel edges.

[3]We draw a random priority for each vertex. The property we require in our analysis for the ordering of the edges is that an edge (u, w) is processed before (u, v) exactly when w's priority is less than v's priority.

2 Technical Overview

Before explaining the techniques used in proving our results, we formalize what we mean for an input $(G, \boldsymbol{x})$ to have vanishing edges values. Our definition includes the Erdős–Rényi random graph with vanishing edge probabilities case, but allows us to work with a sequence of inputs $(G(n), \boldsymbol{x}(n))_{n\geq 1}$ dependent on n in an arbitrary way.

Definition 1. Given an input $(G, \boldsymbol{x}) = (G(n), \boldsymbol{x}(n))$ indexed by a parameter $n \in \mathbb{N}$, we say it has *vanishing edge values*, provided there exists $(\varepsilon(n))_{n\geq 1}$, such that $x_e(n) \leq \varepsilon(n)$ for each edge e of $G(n)$, where $\varepsilon(n) \to 0$ as $n \to \infty$. In this case, we say that $(G(n), \boldsymbol{x}(n))$ has *vanishing edge values with respect to* $\varepsilon(n)$. When the context is clear, we drop the dependence on n in our notation.

In addition, we say a contention resolution scheme is c-selectable on an input $(G, \boldsymbol{x})$ with vanishing edge values, provided it is c_n-selectable on $(G(n), \boldsymbol{x}(n))$, where $c_n \to c$ as $n \to \infty$.

Finally, we say that a sequence of events $(\mathcal{E}(n))_{n\geq 1}$ occurs *with high probability* (w.h.p.), provided $\Pr[\mathcal{E}(n)] \to 1$ as $n \to \infty$.

In the rest of this section, we will summarize the techniques used in proving our results. We begin with our positive results for RCRS and FO-CRS, as the algorithms are very simple to describe, and the analysis is based on the Galton-Watson approach developed by Nuti and Vondrák (2024) for an offline CRS with vanishing edge values. Afterwards, we explain our approach for upper-bounding a FO-CRS, and contrast it with the previously known upper bound of 1/2 for RCRSs. We then discuss our positive result for OCRS, where we first explain why the Galton-Watson approach does not seem applicable in this setting. Finally, we discuss the constructions used in proving our negative results for OCRS, and explain why the arguments differ from the RCRS/FO-CRS case.

RCRS and FO-CRS positive results. Let us assume that $(G, \boldsymbol{x})$, with $G = (V, E)$ is a 1-regular input, i.e., $\sum_{e\in\partial(v)} x_e = 1$ for each $v \in V$. (This is without loss, as we explain in Section 5). Both positive results then analyze the *greedy CRS* on $G = (V, E)$, which accepts any active edge whenever possible, where the only difference between the settings is the order in which the edges are processed. For RCRS, we assume each edge e has an *arrival time* drawn independently and uniformly at random (u.a.r.) from $[0, 1]$. For FO-CRS, we first draw a *seed* from $[0, 1]$ independently and u.a.r for each vertex. We then define $e = (u, v)$ to arrive at time $\min\{t_u, t_v\} + \varepsilon \max\{t_u, t_v\}$, where t_u and t_v are the seeds of u and v, and ε is any number with $\varepsilon < \min_{u\neq v} |t_u - t_v|$. This ensures that the ordering is primarily decided by $\min\{t_u, t_v\}$, and in case there is a tie, the ordering is then decided by $\max\{t_u, t_v\}$. We refer to this order as a *lexicographical* random ordering.

For general edge values, the state-of-the-art is the RCRS of MacRury et al. (2025). This algorithm is very similar to the greedy CRS, except that edges with large x_e are necessarily *dropped* with probability $1 - a(x_e)$ for a carefully engineered function a. Since $a(0) = 1$, this algorithm is asymptotically identical to the greedy CRS for vanishing edge values. Due to this connection, a natural approach to prove Theorem 1.3 would be to improve the analysis of MacRury et al. (2025) for vanishing edge values. Unfortunately, their analysis is inherently lossy—when lower bounding the probability e is matched, it only considers vertices at distance *at most* 3 from u or v.

We follow instead the approach of Nuti and Vondrák (2024) and *reduce* the problem to analyzing a greedy matching algorithm on a certain randomly generated tree. To explain this approach, fix $e = (u, v) \in E$ which is active, and define S_e to be the connected component of e induced by the *active* edges of G. There are three essential observations underlying our analysis:

1. As $n \to \infty$, S_e looks like a (finite) random tree $\mathcal{T}$, constructed by drawing independent Galton-Watson trees $\mathcal{T}_u$ (at u), $\mathcal{T}_v$ (at v) with Poisson parameter 1, and setting $\mathcal{T} = \mathcal{T}_u \cup \mathcal{T}_v \cup \{e\}$.

2. Whether the greedy CRS selects a particular active edge e depends entirely upon S_e, and the order in which the edges of S_e are presented to the algorithm.

3. The edges of S_e are themselves presented to the greedy CRS in a random order.

Using these observations, we equate the probability the greedy CRS selects e to the probability the greedy matching algorithm on $\mathcal{T}$ selects e, when presented the edges of $\mathcal{T}$ in a random order (see Theorem 5.3). Assuming this reduction, let us now sketch how the analysis on $\mathcal{T}$ proceeds for u.a.r. edge arrivals.

We define $q(t)$ to be the probability the vertex u in edge e is not matched when $e = (u, v)$ arrives, given it arrives at time t. Note that due to the symmetry between u and v, this is also the conditional probability v is not matched. Then, since $\mathcal{T}_u$ and $\mathcal{T}_v$ are drawn independently, and e is matched if only if u and v are *not* matched when e arrives, e is matched w.p. $\int_0^1 q(t)^2 dt$ after integrating over the arrival time of e. By considering the children of u in $\mathcal{T}_u$, and using the recursive nature of the Galton-Watson tree process, we then prove that $q(t)$ satisfies a certain integral equation. Using the fact that $q(0) = 1$, we show that the integral equation has the unique solution $q(t) = 1/(t + 1)$, and so $\int_0^1 q(t)^2 dt = 1/2$ as desired.

For a lexicographical random ordering, while the analysis is similar in broad strokes, it is more involved, as we have to consider the function $q(t_1, t_2)$, which is the probability vertex u of e is unmatched, given the seed of u is t_1 and the seed of v is t_2.

FO-CRS negative result. To prove Theorem 1.4, we consider the complete bipartite graph G with $2n$ vertices and uniform edge values on its n^2 edges. We prove the stronger statement, that if a FO-CRS outputs matching $\mathcal{M}$ on G, then $\frac{\mathbb{E}[|\mathcal{M}|]}{n} \leq (1 + o(1)) \left(1 - \ln\left(2 - \frac{1}{e}\right)\right)$ as $n \to \infty$. This is same objective taken for *random-order edge arrivals* by MacRury et al. (2025). For each step $t = 0, \ldots, n^2$, MacRury et al. (2025) track the matching $\mathcal{M}_t$ constructed by an RCRS after the first t random edges are processed in the u.a.r arrival order. Then, they argue that on a typical sample path history $\mathcal{H}_t$, $\mathbb{E}[|\mathcal{M}_{t+1}| - |\mathcal{M}_t| \mid \mathcal{H}_t]$ is upper bounded by roughly $\frac{1}{n}\left(1 - \frac{|\mathcal{M}_t|}{n}\right)^2$, where the expectation is over the endpoints of the $(t + 1)^{th}$ edge, say e_{t+1}, and whether it is active. Viewing $\mathbb{E}[|\mathcal{M}_{t+1}| - |\mathcal{M}_t| \mid \mathcal{H}_t]$ as a "derivative", this allows them to apply the one-sided differential equation method of Bennett and MacRury (2025) to conclude that $|\mathcal{M}_{n^2}|/n$ is roughly at most $w(1) = 1/2$, where w is the unique solution to the differential equation $w'(\tau) = (1 - w(\tau))^2$ with $w(0) = 0$.

The challenge in the the free-order setting is that since the endpoints of e_{t+1} are chosen by the FO-CRS, there is no such simple upper bound on the one step expected change. Instead, we track the "residual subgraph" $R_t \subseteq G$ after the first t edges are processed by the FO-CRS, which contains the remaining edges of G which are compatible with its current matching $\mathcal{M}_t$. Then, we define a "surplus function" $\mathcal{S}$, which assigns a non-negative value $\mathcal{S}(R_t) \geq 0$ to R_t. Intuitively, $\mathcal{S}(R_t)$ is a potential function which is a proxy for the expected matching size available to an FO-CRS which starts with the subgraph $R_t \subseteq G$. We then consider the random variable $Q_t := |\mathcal{M}_t| + \mathcal{S}(R_t)$, and argue that for an appropriately chosen $\mathcal{S}$, this sequence of random variables form a supermartingale, which at $t = 0$, takes the value $(1 + o(1)) \left(1 - \ln\left(2 - \frac{1}{e}\right)\right) n$. These two facts allow use to conclude that $\frac{\mathbb{E}[|\mathcal{M}|]}{n} \leq (1 + o(1)) \left(1 - \ln\left(2 - \frac{1}{e}\right)\right)$ as desired (see Lemma 6.2).

The choice of $\mathcal{S}$ is reverse engineered through our target upper bound of $1 - \ln\left(2 - \frac{1}{e}\right)$, and the technical properties required to argue that the Q_t form a supermartingale. Specifically, if each vertex of G is *marked* independently with probability $1/n$, then for a subgraph $H \subseteq G$, we set

$$\mathcal{S}(H) := -n \ln(\Pr[\text{the marked vertices are independent in } H]).$$

It is not hard to see that $\mathcal{S}(G) = -n \ln \left(2(1-1/n)^n - (1-1/n)^{2n}\right) = \left(1 - \ln\left(2 - \frac{1}{\mathrm{e}}\right) + o(1)\right) n$. The remaining properties required of $\mathcal{S}$ are some conditional probability inequalities involving when the marked process is independent in H. We state and motivate them in greater detail in Section 6.

OCRS positive result. The techniques used in the proof of Theorem 1.1 differ substantially from the RCRS/FO-CRS setting. This is because the decisions of our OCRS depend on the active *and* inactive edges of G. As such, property (2) above does *not* hold, and so we cannot reduce the problem to analyzing a matching algorithm on $\mathcal{T}$. The OCRS of Ezra et al. (2022), which provides the state-of-the-art lower bound on OCRS selectability (MacRury et al., 2025), also has this limitation.

Before we describe our OCRS, we first review the OCRS of Ezra et al. (2022), and explain why it is challenging to prove it is $\alpha \approx 0.382$-selectable on inputs with vanishing edge values (even if we abandon the Galton-Watson approach). This OCRS is given a graph $G = (V, E)$ with an arbitrary fractional matching $\boldsymbol{x} = (x_e)_{e \in E}$. It then selects arriving edge e with probability $c/\Pr[u \text{ and } v \text{ unmatched before } e]$, when e is active and its endpoints are unmatched. This ensures it selects each edge $e \in E$ with probability *exactly* cx_e, and the goal is to make c as large as possible.

In order to prove this is well-defined, one must show $c \leq \Pr[u \text{ and } v \text{ unmatched before } e]$, while assuming the induction hypothesis that every previous edge f before e was selected w.p. cx_f. Now, if G is a tree, then the events "u is matched before e" and "v is matched before e" are independent, and so the induction hypothesis yields $\Pr[u \text{ unmatched before } e] \cdot \Pr[v \text{ unmatched before } e] \geq (1-c)^2$, so one can set $c = \alpha$, as $\alpha = (1-\alpha)^2$.

Suppose now that $(G, \boldsymbol{x}) = (G(n), \boldsymbol{x}(n))$ has vanishing edges values. Nuti and Vondrák (2024) proved that S_e is cycle-free w.h.p (here S_e is defined as before). Thus, restricted to the active edges of G, the neighborhood of e looks *tree-like*. Suppose that conditional on S_e being cycle-free, the events "u is unmatched before e" and "v is unmatched before e" are independent. Since S_e occurs w.h.p., this would be enough to prove $\Pr[u \text{ unmatched before } e] \cdot \Pr[v \text{ unmatched before } e] \geq (1 - o(1))(1-c)^2$, and thus prove Theorem 1.1. However, this conditional independence does *not* hold for arbitrary inputs[4], and so we do not analyze this OCRS.

We instead prove Theorem 1.1 in two self-contained parts: Theorem 3.1 and Theorem 3.2. In Theorem 3.1, we allow $\boldsymbol{x}$ of $G = (V, E)$ to be an arbitrary fractional matching. We say that an edge e is *ℓ-locally-tree-like*, provided there are no cycles in the ℓ-neighborhood of e. We design an OCRS (Algorithm 1) for G in which each active edge which is ℓ-locally-tree-like is matched with probability α_ℓ, where α_ℓ converges to $\alpha \approx 0.382$ as $\ell \to \infty$ (see (3.2) for the explicit definition of α_ℓ). Note that if G has girth $g \geq 5$, then any edge is $\lfloor (g-3)/2 \rfloor$ locally-tree-like, and so Algorithm 1 is $\alpha_{\lfloor (g-3)/2 \rfloor}$-selectable on such inputs. The convergence $\alpha_\ell \to \alpha$ occurs rapidly; in particular, $\alpha_7 \geq 0.3561$, and so Algorithm 1 beats the state-of-the-art of 0.344 for inputs with girth $g \geq 17$.

Algorithm 1 is related to the OCRS of Ezra et al. (2022), however it is *not* defined recursively, and so it is easier to analyze. Specifically, when an active edge $e = (u, v)$ arrives and u, v are unmatched, e is added to the matching with probability $\frac{\alpha}{(1-\alpha x_u(e))(1-\alpha x_v(e))}$, where $x_v(e)$ (respectively, $x_u(e)$) is the fractional degree of u (respectively, v) prior to the arrival of e. Note that if G is a tree, then Algorithm 1 is identical to the OCRS of Ezra et al. (2022), and so it is exactly α-selectable.

The core of the analysis is a *correlation decay* argument in terms of ℓ. We argue that since e is ℓ-locally-tree-like, there is no *short* path between u in v in $G \setminus e$, and so the events "u is matched

[4]Consider a graph on 3 vertices u, v, w with uniform edge values, where (u, v) arrives last. Then $\Pr[u \text{ and } v \text{ matched before } (u,v)] = 0$, yet $\Pr[S_{u,v} \text{ is cycle-free}] > 0$, and $\Pr[u \text{ matched before } e | S_e \text{ cycle-free}]$, $\Pr[v \text{ matched before } e | S_e \text{ cycle-free}] > 0$.

before e" and "v is matched before e" are *asymptotically* independent as $\ell \to \infty$. The analysis here is similar to the online fractional rounding algorithm of Kulkarni et al. (2022), with the main difference being that our analysis works for a general matching $\boldsymbol{x}$, whereas theirs requires $\boldsymbol{x}$ to have vanishing edge values (and in fact be uniform).

The second step of the proof is Theorem 3.2, which provides a *black-box reduction* from the *vanishing edge value regime* to the *locally-tree-like regime*. That is, given a CRS ψ_ℓ which is c_ℓ-selectable for ℓ-locally-tree-like edges, we show how to use it to design a new CRS which is c-selectable for inputs with edge values which are vanishing with respect to ε, where $c = \lim_{\ell\to\infty} c_\ell$. The reduction proceeds via a "two-round exposure" argument, where we split the randomness of each edge state X_e into two random variables, and analyze each separately. We first set $y_e := \log(1/\varepsilon)x_e$ and $z_e := 1/\log(1/\varepsilon)$ for each $e \in E$. Since $\varepsilon = o(1)$, $y_e, z_e \in [0,1]$, and so we can sample $Y_e \sim \mathrm{Ber}(y_e)$ and $Z_e \sim \mathrm{Ber}(z_e)$ for each $e \in E$, such that $X_e = Y_e Z_e$, and for which $((Y_e)_{e\in E}, (Z_e)_{e\in E})$ are independent (see Proposition 3.6).

In the first round, we expose the $(Y_e)_{e\in E}$ random variables, and consider the sub-graph G' of G with edge-set $E' = \{e \in E : Y_e = 1\}$. We argue that for a specific choice of $\ell \to \infty$, any fixed edge of G' is ℓ-locally-tree-like w.h.p. (see Lemma 3.10). At this point, we'd like to expose the $(Z_e)_{e\in E'}$ random variables and execute ψ_ℓ on $(G', (z_e)_{e\in E'})$. However, $(z_e)_{e\in E'}$ may not be a fractional matching of G'. Thus, we first greedily process the edges of G', and create a fractional matching $\boldsymbol{z}' = (z'_e)_{e\in E'}$, where $z'_e \in \{0, z_e\}$ for each $e \in E'$. In Lemma 3.9, we argue that for any fixed edge $e \in E'$, w.h.p. $z'_e = 1/\log(1/\varepsilon)$ (i.e., z'_e will *not* be rounded down from z_e). Let us say an edge $e \in E'$ is *well-behaved*, provided e is ℓ-locally-tree-like, and $z'_e = 1/\log(1/\varepsilon)$. In the second round, we expose the $(Z_e)_{e\in E'}$ random variables, and execute ψ_ℓ on $(G', (z'_e)_{e\in E'})$ to ensure that each well-behaved edge $e \in E'$ is selected with probability $c_\ell z_e$. Since $e \in E$ is an edge of G' with probability y_e, and w.h.p. $e \in E'$ is well-behaved, this ensures that e is matched with an overall probability of $(1 - o(1))c_\ell z_e y_e = (1 - o(1))cx_e$ after averaging over G'.

In order to execute our reduction online, we need to be able to compute $(z'_e)_{e\in E'}$ online. While this can't in general be done for an arbitrary OCRS ψ_ℓ, it can be done if ψ_ℓ is *strongly online*. Roughly speaking, the OCRS should work even if the fractional values of the matching are adversarially revealed. Algorithm 1 of Theorem 3.1 is strongly online, and so Theorem 3.1 and Theorem 3.2 together imply Theorem 1.1.

OCRS negative result. We prove the various results of Theorem 1.2 by considering explicit graphs on which any OCRS cannot perform well. Unlike our upper bound for FO-CRSs, we cannot just upper bound the expected size of the matching output. Instead, we must carefully construct graphs for which no matter the decisions of the OCRS, there exists some edge which cannot be taken with too high a conditional probability.

For part (2) of the theorem, consider a single vertex u and suppose it has edges $\{(u, v_i)\}_{i=1}^n$ around it, with $x_{u,v_i} \approx 1/n$ for all i. For each $i \in [n]$, add a set W_i of $n-1$ children to v_i, with $x_{v_i,w} \approx 1/n$ for each $w \in W_i$. Finally, we add a complete graph of edges amongst the v_i with tiny edge values.

All the edges (v_i, w), $w \in \cup W_i$ are presented to the OCRS first. We say that $A_{v_i} = 1$ if v_i is unmatched after these initial edges are presented to the OCRS. The tiny edges are presented to the OCRS *after* all the edges of the first phase but *before* the edges (u, v_i). The edges (u, v_i) are presented to the OCRS last.

We consider two cases, depending on the size of the minimum pairwise covariance amongst the A_{v_i} (see Lemmas 4.2 and 4.3). Suppose first that after the first phase there is a pair A_{v_i} and A_{v_j} which are not very positively correlated with each other. Then, examining the newly added tiny

edge between v_i and v_j tells us that no OCRS can perform particularly well. Indeed, if the scheme was c-selectable,

$$c \leq \Pr[(v_i, v_j) \in \mathcal{M} \mid X_{v_i,v_j} = 1] \leq \Pr[A_{v_i} = 1, A_{v_j} = 1] = \mathbb{E}[A_{v_i}]\mathbb{E}[A_{v_j}] + \mathrm{Cov}[A_{v_i}, A_{v_j}] \\ \approx (1-c)^2 + \text{something small.}$$

Otherwise, let us suppose all the pairs A_{v_i} and A_{v_j} are positively correlated with each other, and we consider the last edge (u, v_n) that is presented to the algorithm. Note first that it must be the case that if $A_{v_n} = 0$, then a higher than average number of the other A_{v_i} are also zero. Conversely, a fewer than average number of the A_{v_i} are one, since all the random variables are positively correlated.

Suppose that the OCRS does not impose any strong positive correlation between A_{v_n} and the event $\mathcal{E}$ that u is unmatched prior to observing (u, v_n). In this case, exactly like the argument we just made, if the scheme was c-selectable, then

$$c \leq \Pr[\mathcal{E}, A_{v_n} = 1 \mid X_{u,v_n} = 1] = \Pr[\mathcal{E}]\mathbb{E}[A_{v_n}] \approx (1-c)^2 + \text{something small.} \tag{2.1}$$

So the OCRS must try to correlate A_{v_n} with $\mathcal{E}$. But there is a limit to how positively correlated $A_{v_n} = 0$ can be with the event $\neg\mathcal{E}$, because $\neg\mathcal{E}$ can only occur if one of the edges (u, v_i) with $A_{v_i} = 1$ is actually active, and there are a fewer than average number of v_i with this property in the case we are considering.

Next, consider part (3) of the theorem. First, we would like to work with a tree, so we eliminate all the tiny edges we previous had between the v_i. But we also want to obtain an impossibility bound of exactly α. So we need a new mechanism to ensure that A_{v_n} (or perhaps some variation on A_{v_n}) has essentially no positive correlation with $\mathcal{E}$ (or perhaps some variation on $\mathcal{E}$).

We thus consider a *concentrated* OCRS. An OCRS is said to be *concentrated* on an input $(G, \boldsymbol{x})$, provided the matching $\mathcal{M}$ on $(G, \boldsymbol{x})$ it returns satisfies $||\mathcal{M}| - \mathbb{E}[|\mathcal{M}|]| = o(\mathbb{E}[|\mathcal{M}|)$ w.h.p. If the OCRS is concentrated, it must be the case that $\sum A_{v_i}$, while random, still exhibits concentration.

It follows then that we can find a subset S of size $m = o(n)$ such that $\sum_{v_i \in S} A_{v_i}$ is also concentrated. We prove this by taking a uniformly at random subset of size m, and arguing that there's a non-zero chance it has small variance. Now, suppose that all these edges (u, v_i) with $v_i \in S$ are presented to the OCRS last, and we imagine treating all of the edges as if they were a single 'mega-edge'. It is impossible for the OCRS to strongly correlate u being matched prior to this mega-edge being presented to the algorithm with $\sum_{v_i \in S} A_{v_i}$, since the latter sum is approximately deterministic. We use this fact to argue that any concentrated OCRS is, in fact, at most α-selectable.

Finally, to establish part (1) of the theorem, we start by arguing that there is a rearrangement of the v_i so that A_{v_n} is positively correlated with $\sum A_{v_i}$. Then, it must be the case that given $A_{v_n} = 0$, a higher than average number of the other A_{v_i} are also zero. The rest of the argument is then very similar to the argument we used to prove (2.1) in part (2) of the theorem.

3 OCRS Positive Result

As discussed in Section 2, there are two separate self-contained parts to the proof of Theorem 1.1. Let $\alpha \in (0, 1)$ be the unique solution to the equation

$$\alpha = (1-\alpha)^2, \tag{3.1}$$

where $\alpha \approx 0.382$. For each $\ell \geq 1$, we define

$$\alpha_\ell := \left(1 - (1-\alpha)^\ell\right)^2 \alpha, \tag{3.2}$$

which we observe satisfies $\alpha_\ell \to \alpha$ as $\ell \to \infty$. Recall that for an edge $e = (u, v)$, we define the ℓ-neighborhood of e to be those vertices of G at graph distance $\leq \ell$ from u or v, which we denote by $N^\ell(e)$. We also define $D^0(e) := N^0(e)$ and $D^i(e) := N^i(e) \setminus N^{i-1}(e)$ for each $i \geq 1$. We prove that if $N^\ell(e)$ is cycle-free, then e is matched by Algorithm 1 with probability at least $\alpha_\ell x_e$. This motivates the following definition:

Definition 2. Suppose that $\ell \in \mathbb{N}$, $c_\ell \in [0, 1]$, and ψ is a CRS for the matching polytope, whose matching on $(G, \boldsymbol{x})$ we denote by $\psi(G, \boldsymbol{x})$. We say that ψ is *c_ℓ-selectable for ℓ-locally-tree-like edges*, provided the following holds for any input $(G, \boldsymbol{x})$ where $G = (V, E)$:

- For each $e \in E$, if $N^\ell(e)$ is cycle-free, then $\Pr[e \in \psi(G, \boldsymbol{x}) \mid X_e = 1] \geq c_\ell$.

Theorem 3.1. *For each $\ell \geq 1$, Algorithm 1 is α_ℓ-selectable for ℓ-locally-tree-like edges, where α_ℓ is defined in* (3.2)*.*

In order to state our reduction for the second part of the proof of Theorem 1.1, we require the OCRS to satisfy an additional property: When processing an edge e, the decision to match e depends on only (x_e, X_e), as well as the previous edges' fractional values and states (as well as any potential internal randomization used). Thus, the OCRS initially only knows V of G, and learns both X_e *and* x_e of each edge e in an adversarial order. This is in contrast to the usual definition of an OCRS, where it is assumed that the edges of G and their edges values are known upfront. Note that we still work with an oblivious adversary who decides on the arrival order of E prior to drawing $(X_e)_{e \in E}$. We refer to an OCRS with this property as *strongly online*.[5] With this definition, we can now state our reduction.

Theorem 3.2. *Suppose $(\psi_\ell)_{\ell \geq 1}$ is a sequence of CRSs, where each ψ_ℓ is c_ℓ-selectable on ℓ-locally-tree-like edges, and $c := \lim_{\ell \to \infty} c_\ell$. The following then hold:*

1. *There exists an offline CRS ψ which is c-selectable on graphs with vanishing edge values.*
2. *If each ψ_ℓ is strongly online, then ψ can be implemented as an OCRS.*

Since Algorithm 1 is a strongly online CRS, and $\alpha = \lim_\ell \alpha_\ell$, Theorems 3.1 and 3.2 together imply Theorem 1.1. The rest of the section is devoted to proving each individual theorem.

3.1 OCRS for Locally-Tree-Like Edges: Proving Theorem 3.1

Given $G = (V, E)$ with fractional matching $\boldsymbol{x} = (x_e)_{e \in E}$, assume the edges E of G arrive in some adversarial order, where $e' < e$ indicates e' arrives before e. Let $x_v(e) := \sum_{\substack{f < e: \\ f \in \partial(v)}} x_f$ for each $v \in V$ and $e \in \partial(v)$. Observe that $x_v(e)$ is the fractional degree of v, prior to the arrival of e.

[5] The OCRS of Ezra et al. is also strongly online. We do not know whether there is a separation between strongly online OCRSs and usual OCRSs.

Algorithm 1 Tree-OCRS

Input: $G = (V, E)$, $\boldsymbol{x} = (x_e)_{e \in E}$, and α which satisfies (3.1).
Output: subset of active edges forming a matching $\mathcal{M}$

$\mathcal{M} \leftarrow \emptyset$
for arriving $e = (u, v) \in E$ with edge state X_e **do**
    Draw $A_e \sim \text{Ber}\left(\frac{\alpha}{(1-\alpha x_u(e))(1-\alpha x_v(e))}\right)$ independently.
    if u and v are unmatched, $X_e = 1$ and $A_e = 1$ **then**
        $\mathcal{M} \leftarrow \mathcal{M} \cup \{(u, v)\}$
return $\mathcal{M}$

Remark 3.3. Algorithm 1 is well-defined, as $\max\{x_v(e), x_u(e)\} \leq 1$, and $\alpha/(1-\alpha)^2 = 1$ by (3.1). This is the only place we use the specific value of α. Algorithm 1 is well-defined if it was executed with $c \leq \alpha$ instead of α.

In order to prove Theorem 3.1, we must show that if $\mathcal{M}$ is returned by Algorithm 1, then for each $e \in E$ with $N^\ell(e)$ cycle-free,

$$\Pr[e \in \mathcal{M} \mid X_e = 1] \geq \alpha_\ell = \left(1 - (1-\alpha)^\ell\right)^2 \alpha. \tag{3.3}$$

Fix $e = (u, v)$, and let T be the graph induced by the vertices in the ℓ-neighborhood of e. Note that by assumption, T is cycle-free and connected, and so a tree. Denote by ∂T the edges between $D^\ell(e)$ and $D^{\ell+1}(e)$, which we call the *boundary* of T. Let us assume it is non-empty, since otherwise, the neighborhood of e is a tree, and there is nothing to prove.

One difficulty with establishing guarantees on the selectability of an OCRS is the potentially arbitrary order in which edges might be presented to the algorithm. However, when studying *one particular edge* e, we may assume that the edges appear in a much more specifc order. We begin with a preliminary lemma (closely related to the 'tree-matching game' discussed in Kulkarni et al. (2022)) that demonstrates that this is the case:

Lemma 3.4 (proven in §A.1)**.** *When lower bounding* $\Pr[e \in \mathcal{M} \mid X_e = 1]$*, we may assume that the edges of* $T \cup \partial T$ *are processed "bottom up": The edges between* $D^i(e)$ *and* $D^{i+1}(e)$ *are processed before the edges between* $D^{i-1}(e)$ *and* $D^i(e)$ *for* $i = 1, 2, \ldots, \ell$*.*

For the remainder of the proof, we assume that the order of the edges in T and $\partial(T)$ is as described in Lemma 3.4. This lemma is immensely useful, because it lets us come up with a recurrence letting us calculate the value of $\Pr[e \in \mathcal{M} \mid X_e = 1]$. In order to establish this recurrence, first, consider vertices $s \in D^\ell(e)$, and define the random variables Q_s, with $Q_s = 1$ if the OCRS (after making decision on the edges in $\partial(T)$) chooses to leave s unmatched, and 0 otherwise.

Suppose we fix $r \in T$, $r \notin D^\ell(e)$ whose children (in the order they will be processed) are $r_1, \ldots, r_k$ for $k \geq 0$ (if $k = 0$, then r has no children). We can explicitly write out the probability that a vertex r is unmatched after $e_1 = (r, r_1), \ldots, e_k = (r, r_k)$ are processed, conditional on the values of Q_s. Denote this probability by q_r. The key point is that since T is a tree, and edges are processed in a bottom up fashion, once we condition on any particular values of Q_s, the children

$r_1, \ldots, r_k$ of r are matched with their children independently. Thus,

$$\begin{aligned}
q_r &= \prod_{i=1}^{k} \Pr[e_i \notin \mathcal{M} \mid e_j \notin \mathcal{M}, \forall j < i] \\
&= \prod_{i=1}^{k} (1 - \Pr[e_i \in \mathcal{M} \mid e_j \notin \mathcal{M}, \forall j < i]) \\
&= \prod_{i=1}^{k} (1 - \Pr[r_i \text{ is not matched with any of its children, } X_{e_i} = 1, A_{e_i} = 1 \mid e_j \notin \mathcal{M}, \forall j < i]) \\
&= \prod_{i=1}^{k} (1 - \Pr[r_i \text{ is not matched with any of its children, } X_{e_i} = 1, A_{e_i} = 1]) \\
&= \prod_{i=1}^{k} (1 - \Pr[r_i \text{ is not matched with any of its children}] \Pr[X_{e_i} = 1] \Pr[A_{e_i} = 1]) \\
&= \prod_{i=1}^{k} \left(1 - \frac{q_{r_i} x_{e_i} \alpha}{(1 - \alpha x_r(e_i))(1 - \alpha x_{r_i}(e_i))}\right).
\end{aligned}$$

So,

$$q_r = \prod_{i=1}^{k} \left(1 - \frac{q_{r_i} x_{e_i} \alpha}{(1 - \alpha x_r(e_i))(1 - \alpha x_{r_i}(e_i))}\right), \tag{3.4}$$

where $x_r(e_i) = \sum_{j=1}^{i-1} x_{r,r_j}$. If r has no children, then $q_r = 1$. We implicitly understand in the above expression that when we write q_s for $s \in D^\ell(e)$, we actually mean Q_s.

Let $x_r = \sum_{j=1}^{k} x_{r,r_j}$ for convenience, where $x_r = 0$ if $k = 0$ (as x_r never includes the parent of r). We wish to show that if r is "high-enough" up the tree, then $q_r \approx (1 - x_r\alpha)$. To this end, define $\varepsilon_r = 1 - \frac{q_r}{1-\alpha x_r}$. Our goal is to convert the recurrence (3.4) on the q_r into a recurrence on the ε_r. This proceeds similarly to Kulkarni et al. (2022), though we include the derivation for completeness. We first use the equation $q_{r_i} = (1 - \varepsilon_{r_i})(1 - \alpha x_{r_i})$ for $i = 1, \ldots, k$. By applying this to the right-hand of (3.4),

$$\begin{aligned}
q_r = \prod_{i=1}^{k} \left(1 - \frac{q_{r_i} \alpha x_{e_i}}{(1 - \alpha x_r(e_i))(1 - \alpha x_{r_i}(e_i))}\right) &= \prod_{i=1}^{k} \left(1 - \frac{(1 - \varepsilon_{r_i})(1 - \alpha x_{r_i}) \alpha x_{e_i}}{(1 - \alpha x_r(e_i))(1 - \alpha x_{r_i}(e_i))}\right) \\
&= \prod_{i=1}^{k} \left(1 - (1 - \varepsilon_{r_i}) \frac{\alpha x_{e_i}}{(1 - \alpha x_r(e_i))}\right) \\
&= \prod_{i=1}^{k} \left(\frac{1 - \alpha x_r(e_i) - \alpha x_{e_i} + \varepsilon_{r_i} \alpha x_{e_i}}{1 - \alpha x_r(e_i)}\right) \\
&= \prod_{i=1}^{k} \left(\frac{1 - \alpha \sum_{j \le i} x_{r,r_j} + \varepsilon_{r_i} \alpha x_{e_i}}{1 - \alpha \sum_{j < i} x_{r,r_j}}\right) \\
&= \prod_{i=1}^{k} \frac{1 - \alpha \sum_{j \le i} x_{r,r_j}}{1 - \alpha \sum_{j < i} x_{r,r_j}} \left(1 + \frac{\varepsilon_{r_i} \alpha x_{e_i}}{1 - \alpha \sum_{j \le i} x_{r,r_j}}\right) \\
&= (1 - \alpha x_r) \prod_{i=1}^{k} \left(1 + \frac{\varepsilon_{r_i} \alpha x_{e_i}}{1 - \alpha \sum_{j \le i} x_{r,r_j}}\right).
\end{aligned}$$

Thus, applied to (3.4), since $\varepsilon_r = 1 - \frac{q_r}{1-\alpha x_r}$, we get that

$$\varepsilon_r = 1 - \prod_{i=1}^{k} \left(1 + \frac{\varepsilon_{r_i} \alpha x_{e_i}}{1 - \alpha \sum_{j \le i} x_{r,r_j}}\right). \tag{3.5}$$

Note that for $r \in D^\ell(e)$, since $q_r \in \{0,1\}$, $\varepsilon_r = 1$ if $q_r = 0$, and $\varepsilon_r = -\alpha x_r/(1-\alpha x_r) \ge -(1-\alpha)$ if $q_r = 1$. Finally, $\varepsilon_r = 0$ if r has no children.

Given $0 \le j \le \ell$, define the *maximum error* at distance $\ell - j$ from e by $\varepsilon_j^{\max} := \max_{r \in D^{\ell-j}(e)} |\varepsilon_r|$. We argue that the maximum error *decreases* the larger j becomes (and so higher up the tree we go).

Lemma 3.5 (proven in §A.2)**.** *For each $0 \le j \le \ell - 1$, $\varepsilon_{j+1}^{\max} \le \varepsilon_j^{\max}(1-\alpha)$. Thus, $\varepsilon_\ell^{\max} \le (1-\alpha)^\ell$, since $\varepsilon_0^{\max} \le 1$.*

Proof of Theorem 3.1. It suffices to prove (3.3). Fix any values for $(Q_s)_s$. First observe that

$$\Pr[e \in \mathcal{M} \mid X_e = 1, (Q_s)_s] = \Pr[u \text{ and } v \text{ unmatched when } e \text{ arrives}| \ (Q_s)_s] \frac{\alpha}{(1-\alpha x_u(e))(1-\alpha x_v(e))}.$$

Conditional on $(Q_s)_s$, the events that u and v are unmatched are independent, so the first probability equals $q_u q_v$. But $q_u = (1-\alpha x_u(e))(1-\varepsilon_u)$ and $q_v = (1-\alpha x_v(e))(1-\varepsilon_v)$. Thus, since $\max\{\varepsilon_u, \varepsilon_v\} \le (1-\alpha)^\ell$ by Lemma 3.5, it follows that $\Pr[e \in \mathcal{M} \mid X_e = 1, (Q_s)_s] \ge (1-(1-\alpha)^\ell)^2 \alpha$ after cancellation. Averaging over $(Q_s)_s$ proves (3.3). The proof of Theorem 3.1 is thus complete. □

3.2 Reduction to Locally-Tree-Like Edges: Proving Theorem 3.2

We shall make use of the following elementary coupling.

Proposition 3.6 (proven in §A.3)**.** *Suppose $x, y, z \in (0,1)$ satisfy $x \ge yz$. There exists a coupling of Bernoulli random variables $X \sim \mathrm{Ber}(x)$, $Z \sim \mathrm{Ber}(z)$, and $Y \sim \mathrm{Ber}(y)$, where Y, Z are independent, and $X \ge Y \cdot Z$.*

Suppose that $(G, \boldsymbol{x}) = (G(n), \boldsymbol{x}(n))$ satisfies $x_e \le \varepsilon$ for all $e \in E$, where $\varepsilon = \varepsilon(n)$ satisfies $\varepsilon = o(1)$. Fix $\ell = \ell(n)$ where $\ell(n) = \log\log(1/\varepsilon)$. Observe that $\ell(n) \to \infty$ as $n \to \infty$, since $\varepsilon = o(1)$. We describe a CRS ψ for $(G, \boldsymbol{x})$ which makes use of the CRS ψ_ℓ assumed in Theorem 3.2. The idea is to first write $y_e = (1 - 1/\log^{1/4}(\varepsilon^{-1}))x_e \log(1/\varepsilon)$ and $z_e = 1/\log(1/\varepsilon)$, where we observe that $x_e \ge y_e z_e$ for each e. There are then 3 steps to ψ, which we describe below:

1. Sample $G' \subseteq G$: Apply Proposition 3.6 to each $e \in E$, to get $(Y_e)_{e \in E}$ and $(Z_e)_{e \in E}$ such that $X_e \ge Y_e Z_e$, and for which $((Y_e)_{e \in E}, (Z_e)_{e \in E})$ are independent. Let $G' = (V, E')$ be the subgraph of G where $E' := \{e \in E : Y_e = 1\}$.

2. Compute a fractional matching $(z'_e)_{e \in E}$ of G supported on E': Initialize $z'_e := 0$ for each $e \in E$. Then, process the edges $e = (u, v)$ of E in an arbitrary order (such as the one presented by an oblivious adversary if we wish ψ to be an OCRS), and update $z'_e \leftarrow 1/\log(1/\varepsilon)$, if the (current) fractional matching constraints on u and v are not violated. (See Algorithm 2 below for a formal description of this step).

3. Run ψ_ℓ on $(G', \boldsymbol{z}')$ where $\boldsymbol{z}' = (z'_e)_{e \in E'}$: For each $e \in E$, define $Z'_e = Z_e \cdot \mathbf{1}_{[z'_e > 0]}$. Execute ψ_ℓ on $(G', \boldsymbol{z}')$ using $(Z'_e)_{e \in E'}$, and return the matching $\psi_\ell(G', \boldsymbol{z}')$.

Let us denote $\mathcal{M} = \psi_\ell(G', \boldsymbol{z}')$ to be the matching output by ψ. First observe that $\mathcal{M}$ only contains active edges of G, and so ψ is indeed a valid CRS. To see this, observe that a necessary condition for $e \in \mathcal{M}$ is that $Y_e Z'_e = 1$. But since $X_e \geq Y_e Z_e$ by Proposition 3.6, and $Z'_e := Z_e \cdot \mathbf{1}_{[z'_e > 0]}$, it follows that $X_e = 1$.

Algorithm 2 Fractional-Matching

Input: $G = (V, E)$ and $\varepsilon \in [0, 1]$.
Output: a fractional matching $\boldsymbol{z}' = (z'_e)_{e \in E}$ of G supported on E'.

1: Set $z'_e = 0$ for all $e \in E$.
2: **for** arriving $e = (u, v) \in E$ in an arbitrary order $<$ **do**
3: **if** $Y_e = 1$, and $\max\{\sum_{\substack{f \in \partial_G(v): \\ f < e}} z'_f, \sum_{\substack{f \in \partial_G(u): \\ f < e}} z'_f\} \leq 1 - \frac{1}{\log(1/\varepsilon)}$ **then**
4: $z'_e \leftarrow \frac{1}{\log(1/\varepsilon)}$.
5: **return** $\boldsymbol{z}' = (z'_e)_{e \in E}$.

Remark 3.7. Algorithm 2 is stated and analyzed for an arbitrary edge ordering. When proving the OCRS part of Theorem 3.2, we will take this to be the one chosen by an oblivious adversary.

To analyze ψ, we prove three lemmas. The first is a technical point that ensures that once we condition on G', the random variables $(Z'_e)_{e \in E}$ are independent. This ensures that we are indeed passing a valid CRS input to ψ_ℓ in step 3. This follows from the fact that $(z'_e)_{e \in E}$ is a function of G', and $((Y_e)_{e \in E}, (Z_e)_{e \in E})$ are independent by construction.

Lemma 3.8 (proven in §A.4)**.** *Conditional on G' (equivalently $(Y_e)_{e \in E}$), the random variables $(Z'_e)_{e \in E}$ are independently distributed with marginals described by $(z'_e)_{e \in E}$.*

We next show that if $Y_e = 1$, then w.h.p. $z'_e = 1/\log(1/\varepsilon)$. In other words, z'_e will typically be rounded up to the desired value, assuming e is an edge of G'.

Lemma 3.9 (proven in §A.5)**.** *For each $e \in E$,* $\Pr[z'_e = 1/\log(1/\varepsilon) \mid Y_e = 1] \geq 1 - 2\exp\left(-\Omega(\log^{1/2}(1/\varepsilon))\right)$.

Define $C_\ell(e)$ to be the event that there *exists* a cycle in the ℓ-neighborhood of e in G'. We next upper bound the probability that $C_\ell(e)$ occurs, conditional on e being in G' (i.e., $Y_e = 1$).

Lemma 3.10 (proven in §A.6)**.** *For each $e \in E$,* $\Pr[C_\ell(e) \mid Y_e = 1] = O(\varepsilon(\log(1/\varepsilon))^{2\ell+1})$.

We are now ready to prove Theorem 3.2.

Proof of Theorem 3.2. In order to analyze ψ on G, fix $e = (u, v) \in E$. Now, $e \in \mathcal{M}$ if and only if $e \in \psi_\ell(G', \boldsymbol{z}')$. We shall first show that

$$\Pr[e \in \psi_\ell(G', \boldsymbol{z}')] \geq (1 - o(1))cx_e, \tag{3.6}$$

Recall that $C_\ell(e)$ is the event that there *exists* a cycle in the ℓ-neighborhood of e in G'. Observe that $C_\ell(e)$ is a function of $(Y_f)_{f \in E}$ (i.e., it is $(Y_f)_{f \in E}$-measurable). Moreover, since ψ_ℓ is c_ℓ-selectable on ℓ-locally-tree-like edges by assumption, we can apply Lemma 3.8 to get that

$$\Pr[e \in \psi_\ell(G', \boldsymbol{z}') \mid (Y_f)_{f \in E \setminus e}, Y_e = 1] \cdot \mathbf{1}_{\neg C_\ell(e)} \geq c_\ell z'_e \cdot \mathbf{1}_{\neg C_\ell(e)}.$$

Thus, after taking expectations over $(Y_f)_{f \in E \setminus e}$,

$$\Pr[\{e \in \psi_\ell(G', \boldsymbol{z}')\} \cap \neg C_\ell(e) \mid Y_e = 1] \geq c_\ell \mathbb{E}[z'_e \cdot \mathbf{1}_{\neg C_\ell(e)} \mid Y_e = 1]. \tag{3.7}$$

We now must lower bound $\mathbb{E}[z'_e \cdot \mathbf{1}_{\neg C_\ell(e)} \mid Y_e = 1]$. By applying Lemmas 3.9 and 3.10, we get that

$$\begin{aligned}\mathbb{E}[z'_e \cdot \mathbf{1}_{\neg C_\ell(e)} \mid Y_e = 1] &\geq \frac{1}{\log(1/\varepsilon)} \left(1 - O(\varepsilon(\log(1/\varepsilon))^{2\ell+1}) - 2\exp\left(-\Omega(\log^{1/2}(1/\varepsilon))\right)\right) \\ &= (1 - o(1))\frac{1}{\log(1/\varepsilon)},\end{aligned}$$

where the final line uses that $\varepsilon(\log(1/\varepsilon))^{2\ell+1}) = o(1)$ for $\ell(n) = \log\log(1/\varepsilon)$. By combining everything together,

$$\Pr[e \in \psi_\ell(G', \boldsymbol{z}') \mid Y_e = 1] \geq \Pr[\{e \in \psi_\ell(G', \boldsymbol{z}')\} \cap \neg C_\ell(e) \mid Y_e = 1] \geq (1 - o(1))\frac{c_\ell}{\log(1/\varepsilon)}.$$

Finally, recalling that $y_e = (1 - 1/\log^{1/4}(\varepsilon^{-1}))x_e \cdot \log(1/\varepsilon)$, we get that

$$\Pr[e \in \psi_\ell(G', \boldsymbol{z}') \mid X_e = 1] \geq (1 - o(1))c_\ell = (1 - o(1))c$$

after cancellation, and using the fact that $c_\ell \to c$ as $n \to \infty$, since $\ell \to \infty$ as $n \to \infty$. Thus, ψ is c-selectable for graphs with vanishing edges values.

To complete the proof, we must argue that if each ψ_ℓ is strongly-online, then ψ can be implemented as an OCRS as the edge states $(X_e)_{e \in E}$ of $(G, \boldsymbol{x})$ are revealed. First observe that ψ is given $(G, \boldsymbol{x})$ in advance, and thus knows ε. This means that steps 1. and 2. (i.e., Algorithm 2) of ψ involving edge $e \in E$ can be implemented when e arrives. In order to execute step 3. online, we pass on matching e if $Y_e = 0$ (i.e., e is not an edge of G'). Otherwise, we make use of the strongly online property of ψ_ℓ: Observe that the values $(z'_f, Z'_f)_{f \leq e, Y_f = 1}$ are known when e arrives, and determine whether or not $e \in \psi_\ell(\boldsymbol{z}', G')$. Thus, when $Y_e = 1$, we can present ψ_ℓ the values $(z'_f, Z'_f)_{f \leq e, Y_f = 1}$ to determine whether to match e. The resulting matching will be identical to the offline setting in which ψ_ℓ is given the entire input $(G', \boldsymbol{z}')$ in advance. □

4 OCRS Negative Results

4.1 General Graphs

We define a sequence of inputs dependent on $n \in \mathbb{N}$ to prove the upper bound. Start with a vertex u whose neighbors $N(u) = \{v_1, \ldots, v_n\}$ have edge values $x_{u,v_i} = 1/n$ for $i \in [n]$. Next, add a clique amongst the vertices of $N(u)$ where $x_{v_i,v_j} = 1/n(n-1)$ for each $i, j \in [n]$, $i \neq j$. Finally, each v_i has its own $n - 2$ additional neighbors, denoted W_i, where $x_{v_i,w} = 1/n$ for each $w \in W_i$ and $i = 1, \ldots, n$. We define $W = \cup_{i=1}^n W_i$ for convenience, and refer to the input by $(G(n), \boldsymbol{x}(n))$, which we denote by $(G, \boldsymbol{x})$ when clear (see Figure 1 for an illustration). We next specify the order in which the edges are presented to the OCRS. This is done in 3 *phases*:

1. Present the edges of $N(u) \times W$ in an arbitrary order.
2. Present the edges between the vertices of $N(u)$ in an arbitrary order.
3. Present the edges of $\partial(u)$ in an arbitrary order.

We refer to an order specified in this way as a *phase-based order*. Our negative result holds for any such order, even if it is presented to the OCRS ahead of time. Let us define $\beta \in (0, 1)$ to be the unique solution to the below equation:

$$0 = 1 - 2\beta - \beta\exp(2 - 1/\beta), \tag{4.1}$$

where $\beta \approx 0.3895$.

Theorem 4.1. *Any OCRS which processes the edges of $(G, \boldsymbol{x})$ in a phase-based order is at most $(\beta + o(1))$-selectable on $(G, \boldsymbol{x})$.*

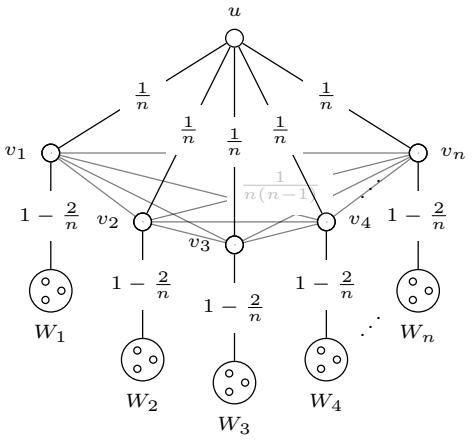


Figure 1: Graph to prove negative result. Each W_i consists of $n-2$ vertices.

In order to prove Theorem 4.1, let us fix an OCRS for $(G, \boldsymbol{x})$ which returns the matching $\mathcal{M}$ when presented the edges in a phase-based order, and which is c-selectable for $c \in (0, 1)$. Our goal is to upper bound c. It is not hard to show that without loss of generality, we may assume that the OCRS is *exactly*[6] c-selectable. That is, $\Pr[e \in \mathcal{M} \mid X_e = 1] = c$ for each edge e of G.

Define M_{v_i} (respectively, $A_{v_i} = 1 - M_{v_i}$) to be an indicator random variable for the event vertex v_i is matched (respectively, not matched) after the first phase. Observe first that for each $i \in [n]$,

$$\mathbb{E}[M_{v_i}] = \Pr[\cup_{w \in W_i} (v_i, w) \in \mathcal{M}] = \sum_{w \in W_i} c x_{v,w} = c\left(1 - \frac{2}{n}\right), \tag{4.2}$$

where the last equality uses that $\sum_{w \in W_i} x_{v_i,w} = 1 - 2/n$. At this point, let us consider the minimum covariance amongst the $(A_{v_i})_{i=1}^n$ random variables. Specifically, define $\theta := \min_{i \neq j} \mathrm{Cov}[A_{v_i}, A_{v_j}]$, where $\mathrm{Cov}[A_{v_i}, A_{v_j}] := \mathbb{E}[A_{v_i} A_{v_j}] - \mathbb{E}[A_{v_i}]\mathbb{E}[A_{v_j}]$. We first lower bound θ as a function of c for $n \to \infty$. This gives a tighter upper bound on c the *smaller* θ is.

Lemma 4.2 (Small covariance regime: proven in §B.1)**.** $\theta \geq (1 + o(1))(c - (1-c)^2)$.

This lemma follows from examining each of the edges (v_i, v_j), and noting that we can only select it if each of its end points is available, so c must be less than $\mathbb{E}[A_{v_i} A_{v_j}]$, which is less than $\mathbb{E}[A_{v_i}]\mathbb{E}[A_{v_j}] + \theta$. We next upper bound a function of c and θ by 1. This leads to a tighter upper bound on c the *larger* θ is.

Lemma 4.3 (Large covariance regime: proven in §B.2)**.** $2c + c\exp(c + \theta/c - 1) \leq 1 + o(1)$.

The main idea behind this lemma is to examine the last edge, consider the probability that both its vertices are already matched when the edge arrives, and show that this probability is not too large. When the A_{v_i} are positively correlated, conditioning on $A_{v_n} = 0$ leaves fewer available

[6]Given a c-selectable OCRS for $(G, \boldsymbol{x})$ which returns the matching $\mathcal{M}$, we can turn it into an exactly c-selectable OCRS by independently *dropping* any selected edge with probability $\Pr[e \in \mathcal{M} \mid X_e = 1] - c$. We then continue to simulate the original OCRS, including its dropped edges, so this reduction can be executed as the edges arrive online. In the concentrated case, we work directly with the original OCRS and do not use this reduction.

neighbors for u to get matched with. If none of the edges to these neighbors is active, there is no way u could be matched before the arrival of the last edge. Considering this activity probability gives the exponential term after applying Jensen's inequality

We are now ready to prove Theorem 4.1. We combine the inequalities from Lemmas 4.2 and 4.3 to determine the optimal choice of θ for the OCRS.

Proof of Theorem 4.1. Recall that β satisfies (4.1). Now, for any OCRS which is exactly c-selectable on $(G, \boldsymbol{x})$, Lemmas 4.2 and 4.3 together imply that

$$2c + c\exp(c + (c - (1-c)^2)/c - 1) \leq 1 + o(1).$$

Thus, $1 - 2c - c\exp(2 - 1/c) \geq o(1)$ after rearranging and simplifying. Now, the function $z \to 1 - 2z - z\exp(2 - 1/z)$ is continuous, and non-negative for $z \leq \beta$, so we may conclude that $c \leq \beta + o(1)$, as required. □

4.2 Trees

In this section, we first establish a negative result for *concentrated* OCRSs on trees. We refer to an OCRS as *concentrated* on $(G, \boldsymbol{x})$, provided the matching $\mathcal{M}$ on $(G, \boldsymbol{x})$ it returns satisfies $||\mathcal{M}| - \mathbb{E}[|\mathcal{M}|]| = o(\mathbb{E}[|\mathcal{M}|])$ w.h.p. As we discuss in Section E, Algorithm 1, the OCRS of Ezra et al. (2022), and the OCRS of Aminian et al. (2026) all satisfy this property. Throughout this section, let us fix a particular concentrated OCRS for $(G, \boldsymbol{x})$ which returns a matching $\mathcal{M}$, and is c-selectable for $c \in (0, 1)$, i.e., $\Pr[e \in \mathcal{M} \mid X_e = 1] = c$ for each edge e of G.

As in the hardness result for general graphs, we begin by defining a sequence of inputs dependent on $n \in \mathbb{N}$ to prove the hardness result. Start with a vertex u whose neighbors $N(u) = \{v_1, \ldots, v_n\}$ have edge values $x_{u,v_i} = 1/n$ for $i \in [n]$. Each v_i has its own $n - 1$ additional neighbors, denoted W_i, where $x_{v_i,w} = 1/n$ for each $w \in W_i$ and $i = 1, \ldots, n$. We refer to this input by $(G(n), \boldsymbol{x}(n))$ (see Figure 2 for an illustration). The edges are presented to the OCRS in 2 phases. In the first

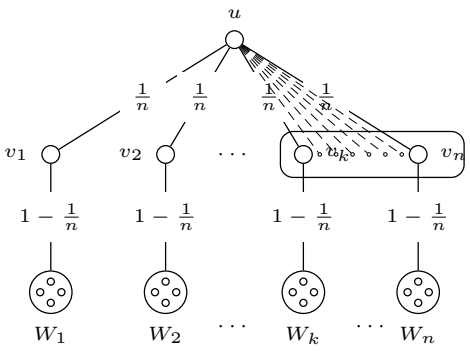


Figure 2: Tree to prove negative result. Each W_i consists of $n - 1$ vertices.

phase, the edges of $N(u) \times \cup_{i=1}^{n} W_i$ are presented in an arbitrary order. At this point, we define M_{v_i} (respectively, $A_{v_i} = 1 - M_{v_i}$) as an indicator random variable for the event that vertex v_i is matched (respectively, not matched) after the first phase. The distribution of $(A_{v_i})_{i=1}^{n}$ is not affected by the arrival order of the second-phase edges. Before we can discuss the order in which the edges of the second phase are presented to the OCRS, we must briefly digress to discuss an important property of a concentrated OCRS.

Since $\sum_{i=1}^{n} M_{v_i}$ and $|\mathcal{M}|$ differ by at most 1 and we've assumed that $|\mathcal{M}|$ is concentrated about $\mathbb{E}[|\mathcal{M}|]$, we know that $\sum_{i=1}^{n} M_{v_i}$ is also concentrated. Thus, by applying elementary bounds, we get

that $\mathrm{Var}[\sum_{i=1}^n M_{v_i}] = o(n^2)$, and so we must have that $\mathrm{Var}[\sum_{i=1}^n A_{v_i}] = o(n^2)$. For concreteness, let us say that $\mathrm{Var}[\sum_{i=1}^n A_{v_i}] \leq \theta_n n^2$, with $\theta_n \to 0$, and $\theta_n \geq \frac{1}{\sqrt{n}}$. Note that since we are only demanding an upper bound on $\mathrm{Var}[\sum_{i=1}^n A_{v_i}]$, we can assume w.l.o.g. $\theta_n \geq \frac{1}{\sqrt{n}}$. This will just be a convenient upper bound to use on the 'true' θ_n in the following calculations.

We have the following probabilistic lemma:

Lemma 4.4 (proven in §B.3)**.** *Suppose $n \to \infty$. Let $A_1, A_2, \ldots, A_n$ be random variables, with $\mathrm{Var}(A_i) < \delta$. Define $A = \sum_{i=1}^n A_i$, and $A_S = \sum_{i \in S} A_i$. Suppose $Var(A) \leq \theta_n n^2$ and $\theta_n \geq \frac{1}{\sqrt{n}}$. Then, we can find $S \subseteq \{1, 2, \ldots, n\}$ with $|S| = m = \theta_n^{1/4} n$ such that $Var(A_S) \leq 3\theta_n m^2$ for sufficiently large n.*

We prove this lemma simply by considering the average value of the variance over all possible sets S of size m. The lemma implies that for $m = \theta_n^{1/4} n$, we can find a set $S \subseteq \{1, 2, \ldots, n\}$ of size m such that $\mathrm{Var}(\sum_{i \in S} A_{v_i}) \leq 3\theta_n m^2$. Let us assume that in phase 2, the edges of $\partial(u)$ are presented in an order that ensures that the edges $\{u\} \times S = \{(u, v_i) : v_i \in S\}$ come last. We emphasize that this is a valid ordering, as the adversary has access to the OCRSs description ahead of time.

Theorem 4.5. *Any concentrated OCRS which processes the edges of $(G, \boldsymbol{x})$ in two phases as described above is at most $(\alpha + o(1))$-selectable, where α is defined in* (3.1)*.*

For convenience, let us immediately relabel the vertices so that the edges of $\partial(u)$ are presented in the order $e_1 = (u, v_1), \ldots, e_n = (u, v_n)$ to the OCRS. We will group the last m vertices from v_k to v_n together ($k = n - m + 1$), and think about them as if they were one last vertex. We do this to ensure that the algorithm's decisions before reaching the last vertex cannot be very correlated with the availability of the last vertex. It will be useful to define:

$$A_{end} = \sum_{i=k}^{n} A_{v_i}, \mathcal{B} = \{\cup_{j=k}^{n} \{X_{e_j} A_{v_j} = 1\}\}, \text{ and } \mathcal{C} = \{M_u(e_k) = 0\}.$$

Here $\mathcal{B}$ represents the event that there is something for the OCRS to select amongst the last m edges, and $\mathcal{C}$ represents the event that the OCRS even reaches the last m vertices, i.e., it has not already made a selection when it reaches e_k. Note that we have ensured that $\mathrm{Var}(A_{end}) = o(m^2)$.

Since the OCRS must select each of the last m edges $e_k, \ldots, e_n$ with probability at least $\frac{c}{n}$, and it can only do this if both the events $\mathcal{B}$ and $\mathcal{C}$ happen, we conclude that:

$$\frac{cm}{n} \leq \Pr[\mathcal{B} \cap \mathcal{C}],$$

We would now like to prove an upper bound on the probability that $\mathcal{B}$ and $\mathcal{C}$ simultaneously occur. Since A_{end} is concentrated, it is approximately deterministic. Therefore, the event $\mathcal{B}$, which is a function of A_{end} and X_{e_j} for $j \geq k$, must necessarily be independent of the algorithm's matching decisions before e_k since these decisions are made without knowledge of X_{e_j} for $j \geq k$. This observation lets us establish following lemma:

Lemma 4.6 (proven in §B.4)**.** *Suppose $\Pr[A_{end} > T] \leq \varepsilon$. Then,*

$$\Pr[\mathcal{B} \cap \mathcal{C}] \leq \frac{T}{n} \Pr[\mathcal{C}] + \varepsilon \leq \frac{T}{n} \left(1 - \frac{c(n-m)}{n}\right) + \varepsilon.$$

We are now ready to prove Theorem 4.5.

Proof of Theorem 4.5. Note that we assumed $m = \theta_n^{1/4} n$. Plugging this into the inequality, and using Lemma 4.6 we get that

$$c\theta_n^{1/4} \le \frac{T}{n}\left(1 - c + c\theta_n^{1/4}\right) + \varepsilon.$$

Since each v_i is matched in the first phase with probability at least $c(n-1)/n$, we have $\mathbb{E}[A_{end}] \le (1-c+c/n)m$. If we let $T = (1-c+c/n)m + \theta_n^{1/4} m$, Chebyshev's inequality tells us that $\Pr[A_{end} > T] \le \frac{3\theta_n m^2}{\theta_n^{1/2} m^2} = 3\theta_n^{1/2}$. Therefore,

$$c \le \left(1 - c + c/n + \theta_n^{1/4}\right)\left(1 - c + c\theta_n^{1/4}\right) + 3\theta_n^{1/4}.$$

Taking $n \to \infty$ yields the desired result. □

We next consider when the OCRS is not necessarily concentrated on $(G, \boldsymbol{x})$. The proof here follows closely to the proof of Theorem 4.1, with the main difference being that we focus on the average covariance of $(A_{v_i})_{i=1}^n$, as opposed to the minimum covariance.

Theorem 4.7 (proven in §B.5)**.** *Any OCRS is at most* $(\gamma + o(1))$*-selectable on* $(G, \boldsymbol{x})$ *where* γ *is the unique solution to the equation:* $0 = 1 - 2\gamma - \gamma\exp(\gamma - 1)$*, and* $\gamma \approx 0.3929$*.*

5 RO-CRS and FO-CRS Positive Results

We now investigate contention resolution schemes that assume that the edges of the input $(G, \boldsymbol{x}) = (G(n), \boldsymbol{x}(n))$ arrive in a random order drawn u.a.r., or in an order chosen by the CRS. In the latter case, the edge-ordering is a *non-uniform* random ordering. For both arrival orders, we study the GREEDY-CRS which matches an arriving edge $e = (u, v)$ provided e is active, and u and v were previously unmatched.

For general inputs with vanishing edge values, Brubach et al. (2024) showed that GREEDY-CRS is $(1 - e^{-2})/2 \approx 0.432$-selectable, and it is not hard to see that their analysis is tight[7]. In order to surpass this bound, one approach is to ensure that $(G, \boldsymbol{x})$ is 1-*regular*; that is, $\sum_{e \in \partial(v)} x_e = 1$ for each $v \in V$. Fu et al. (2021) were the first to prove a reduction to this setting in the related vertex arrival model (see Lemma 2 in Section 3.1). MacRury et al. (2025) applied this reduction for random-order edge arrivals (Lemma 1 in Appendix C), and Nuti and Vondrák (2024) applied it in the offline setting. Nuti and Vondrák (2024) also observed that this reduction preserves vanishing edge values, no matter the arrival order of the edges. We therefore make this assumption throughout the section.

Theorem 5.1. *For* 1*-regular inputs with vanishing edge values,* GREEDY-CRS *is* $\frac{1}{2}$*-selectable when the edges arrive u.a.r.*

Recall that it is impossible for an RCRS to be better than $\frac{1}{2}$-selectable even for vanishing edge values (see Theorem 5 in Section 3.3 in MacRury et al. (2025)). We next show that it is possible to surpass this impossibility result in the free-order setting, where we now analyze GREEDY-CRS in a *non-uniform* random order.

[7] Consider $G = (V, E)$ with $V = \{u_i, v_i\}_{i=0}^n$ and $E = \{(u_0, v_0)\} \cup \{(u_0, u_i), (v_0, v_i)\}_{i=1}^n$ The probability x_e of every $e \in E$ is $1/(n+1)$. When (u_0, v_0) is active, GREEDY-CRS selects (u_0, v_0) with probability *exactly* $(1 - e^{-2})/2$ as $n \to \infty$.

Theorem 5.2. *For* 1*-regular inputs with vanishing edge values,* GREEDY-CRS *is* $1 - \ln\left(2 - \frac{1}{e}\right) \approx 0.510$*-selectable when the edges arrive in a certain non-uniform random order.*

It will be convenient to assume for the proof of Theorem 5.1 that each edge e arrives at a time that is chosen independently and u.a.r. from $[0, 1]$. Furthermore, for the proof of Theorem 5.2, we will assume that we generate for each vertex an independent seed u.a.r. from $[0, 1]$, and that the edges then arrive in lexicographic order, i.e., each edge is associated with the seeds of its vertices $\{x, y\}$, and it arrives at time $\min\{x, y\} + \epsilon \max\{x, y\}$, where ϵ is any number less than the distance between any two vertex seeds. The primary property we will need for our ordering is that an edge (u, w) is processed before (u, v) exactly when w's seed is less than v's seed, so our proof will also work for orderings other than the lexicographic one. In any such case, we note that our analysis of GREEDY-CRS is in fact tight. The proofs of Theorems 5.1 and 5.2 both first reduce to the random tree case, and so we treat them in a unified manner.

5.1 Reduction to the Random Tree Case

This section follows Nuti and Vondrák (2024) very closely; however for completeness we present all the proofs in Section C. Recall that S_e is the connected component of e induced by the active edges of G. Consider a simple method to construct a random tree $\mathcal{T}$ that will estimate S_e as $n \to \infty$. Let u and v be two special vertices connected by an edge, and generate the random tree $\mathcal{T}$ by generating two independent Galton-Watson processes at u and v. Explicitly, construct $\mathcal{T}$ as follows: Start with the two special vertices u and v unmarked and connected by an edge, and add them into $\mathcal{T}$. Then, for each unmarked vertex w added into $\mathcal{T}$, obtain an independent sample L_w of a Poisson random variable with mean 1, add L_w new children to w, and then mark w. End the construction when all vertices of $\mathcal{T}$ are marked. Standard results about Galton-Watson processes immediately tell us that $\mathcal{T}$ is finite with probability 1 (e.g., see Theorem 6.1 in Harris (1964)).

After we generate $\mathcal{T}$, clearly it is possible to run a greedy matching algorithm on $\mathcal{T}$ (either after generating random times of arrival for each edge $e \in \mathcal{T}$, or random seeds for each vertex $w \in \mathcal{T}$ as the case may be). This produces a matching $\mathcal{M}_{\mathcal{T}}$. We claim the following relation between the execution of the greedy CRS on $G(n)$ and the greedy matching algorithm on $\mathcal{T}$:

Theorem 5.3 (proven in §C)**.** *There exists a function* f *satisfying* $\lim_{\varepsilon \to 0} f(\varepsilon) = 0$ *such that for any* 1*-regular input* $(G(n), \boldsymbol{x}(n))$ *with vanishing edge values, and for all* $e \in G(n)$*, we have:*

$$|\Pr[e \in \mathcal{M} \mid X_e = 1] - \Pr[(u, v) \in \mathcal{M}_{\mathcal{T}}]| \leq f(\varepsilon(n)),$$

where $\mathcal{M}$ *is the matching produced by* GREEDY-CRS *when run on* $G(n)$*.*

Clearly, establishing Theorem 5.3 reduces the proofs of Theorems 5.1 and 5.2 to the calculation of $\Pr[(u, v) \in \mathcal{M}_{\mathcal{T}}]$ in two different random orderings of the edges in $\mathcal{T}$.

5.2 Proving Theorems 5.1 and 5.2

Before we begin the proofs of these theorems, we will establish some common notions between the two proofs. First, we say that a vertex $w \in \mathcal{T}$ has been matched if an edge adjacent to it has been matched by the greedy matching algorithm. Second, in order for u to be unmatched when (u, v) arrives, we need every edge (u, w) from u to one of its children w to satisfy:

1. It is processed after edge (u, v), *or*
2. w is already matched when (u, w) is processed.

Let us say that a child w of u is *bad* if it satisfies neither of these criteria, i.e., (u, w) is processed before (u, v) and w is unmatched at this point. Observe that (u, v) is matched by the greedy algorithm precisely when no child of u or v is bad.

Proof of Theorem 5.1. We aim to calculate $\Pr[(u, v) \in \mathcal{M}_{\mathcal{T}}]$ when each edge e of the random tree $\mathcal{T}$ arrives at a time that is chosen independently and uniformly at random from $[0, 1]$.

Let us define for any $x \in [0, 1]$, the function $q(x)$ which denotes the probability that the special vertex u is *unmatched* at the time x when (u, v) arrives. The probability is taken over all possible realizations of $\mathcal{T}$, and all possible arrival times of the edges in $\mathcal{T}$ (other than (u, v)). Note briefly that q is continuous—indeed,

$$|q(x) - q(x + h)| \leq \Pr[\text{some edge arrives in between times } x \text{ and } x + h].$$

The probability that (u, v) is selected by the greedy algorithm is $\int_0^1 q(x)^2 dx$. This is because:

1. The symmetry between u and v ensures $q(x)$ is also the conditional probability that v is unmatched.
2. Whether u or v are matched before the edge (u, v) arrives are independent events.
3. (u, v) is added into the matching exactly if neither u nor v are matched when it arrives.

It turns out that we can actually determine what $q(x)$ is using a recursion. Critically, for an edge (u, w) arriving at time z, the probability (taken over all possible realizations of how the process looks below w) that w is unmatched when (u, w) arrives is exactly $q(z)$, and so, the probability that w is bad is exactly $\int_0^x q(z)dz$. Furthermore, the chance that u has k children is $\frac{e^{-1}}{k!}$, and whether these children are bad are independent events. It follows that

$$\begin{aligned} q(x) = \sum_{k=0}^{\infty} \Pr[u \text{ has } k \text{ children and none of them are bad}] &= \sum_{k=0}^{\infty} \frac{e^{-1}}{k!} \left(1 - \int_0^x q(z)dz\right)^k \\ &= \exp\left(-\int_0^x q(z)dz\right). \end{aligned}$$

To solve this equation, note that q is differentiable in x (since $\int_0^x q(z)dz$ is differentiable in x) and $\frac{dq}{dx} = -q(x)^2$. We know that $q(0) = 1$, so this is a differential equation we can solve using separation of variables. We conclude that $q(x) = \frac{1}{x+1}$. Finally, note that

$$\int_0^1 q(x)^2 dx = -\int_0^1 \frac{dq}{dx} dx = q(0) - q(1) = \frac{1}{2}.$$

This concludes the proof of Theorem 5.1. □

The argument to prove Theorem 5.2 is similar to the previous proof, but we need to be slightly more careful in how we define q.

Proof of Theorem 5.2. Once more, we aim to calculate $\Pr[(u, v) \in \mathcal{M}_{\mathcal{T}}]$, but this time each edge e of the random tree $\mathcal{T}$ arrives at a time determined by the random seeds of the edge's associated vertices.

Let us define for any $x, y \in [0,1]$, the function $q(x,y)$ which denotes the probability that the special vertex u is unmatched when (u,v) arrives given u has the seed x and v has the seed y. The probability that (u,v) is selected by the greedy algorithm is now:

$$\int_0^1 \int_0^1 q(x,y)q(y,x)dxdy.$$

Once again, it turns out that we can establish a recurrence for q. Indeed, for an edge (u,w) in which w has seed z, the probability (taken over all possible realizations of how the process looks below w) that w is unmatched when (u,w) arrives is exactly $q(z,x)$, and so, the probability that (u,w) is bad is exactly $\int_0^y q(z,x)dz$ (since the edge (u,w) is processed before (u,v) exactly if $z \leq y$). It follows that

$$\begin{aligned} q(x,y) &= \sum_{k=0}^{\infty} \Pr[u \text{ has } k \text{ children and none of them are bad}] \\ &= \sum_{k=0}^{\infty} \frac{e^{-1}}{k!} \left(1 - \int_0^y q(z,x)dz\right)^k \\ &= \exp\left(-\int_0^y q(z,x)dz\right). \end{aligned}$$

To solve this equation, note that q is differentiable in y (since $\int_0^y q(z,x)dz$ is differentiable in y) and

$$\frac{\partial}{\partial y} q(x,y) = -q(x,y)q(y,x).$$

Since $q(x,y)$ is differentiable in y, $\int_0^y q(z,x)dz$ is differentiable in x, and so q is differentiable in x as well. Then,

$$\begin{aligned} \frac{\partial}{\partial x} q(x,y) &= q(x,y)\left(-\int_0^y \frac{\partial}{\partial x} q(z,x)dz\right) \\ &= q(x,y)\left(\int_0^y q(z,x)q(x,z)dz\right) \\ &= q(x,y)\left(-\int_0^y \frac{\partial}{\partial z} q(x,z)dz\right) \\ &= q(x,y)\left(1 - q(x,y)\right), \end{aligned}$$

since $q(x,0) = 1$. We know that $q(0,y) = e^{-y}$, so this is a differential equation we can solve for each value of y separately. It follows using variable separation that

$$q(x,y) = \frac{e^x}{e^x + e^y - 1}.$$

Finally, note that

$$\begin{aligned} \int_0^1 \int_0^1 q(x,y)q(y,x)dxdy &= -\int_0^1 \int_0^1 \frac{\partial}{\partial y} q(x,y)dydx \\ &= \int_0^1 q(x,0) - q(x,1)dx \\ &= 1 - \int_0^1 \frac{e^x}{e^x + e - 1}dx \\ &= 1 - \ln\left(2 - \frac{1}{e}\right), \end{aligned}$$

where we computed the last integral by substituting $u = e^x$. $\square$

6 FO-CRS Negative Result

To prove Theorem 1.5, we consider the complete bipartite graph with $2n$ vertices and uniform edge values. Specifically, let $G = (U_1, U_2, U_1 \times U_2)$ where $|U_1| = |U_2| = n$ for $n \geq 1$, and set $x_e = p = 1/n$ for all $e \in U_1 \times U_2$. We denote this input by $(G, \boldsymbol{x})$, implicitly dropping the dependence on n. For $U \subseteq U_1 \cup U_2$, we denote $G[U]$ as the subgraph induced by vertices U. Furthermore, for a subgraph $H \subseteq G$, let $V(H)$ denote its vertices, and $E(H)$ denote its edges. Finally, for each $F \subseteq E(H)$ and $U \subseteq V(H)$, let $H \setminus F$ be H with the edges of F removed, and $H - U$ be H with the vertices of U removed, including all of their incident edges in H. Throughout the section, all of our asymptotics are with respect to $n \to \infty$.

Lemma 6.1. *For any FO-CRS which outputs matching $\mathcal{M}$ on $(G, \boldsymbol{x})$,*

$$\frac{\mathbb{E}[|\mathcal{M}|]}{n} \leq (1 + o(1)) \left(1 - \ln\left(2 - \frac{1}{e}\right)\right).$$

Lemma 6.1 immediately implies Theorem 1.4, so we focus on establishing it for the rest of the section. We note that any FO-CRS which maximizes $\mathbb{E}[|\mathcal{M}|]$ can be assumed to be deterministic without loss (by averaging over its internal random bits), and so we hereby make this assumption.

For each $1 \leq t \leq n^2$, let e_t be the tth edge of G processed by the FO-CRS, and denote its state by X_{e_t} (clearly, $X_{e_t} \sim \text{Ber}(1/n)$). Let $\mathcal{M}_t$ denote the matching constructed by the FO-CRS after t rounds, and F_t denote the *free vertices* of G which are unmatched by $\mathcal{M}_t$. Since the FO-CRS is deterministic, $\mathcal{M}_t$ and F_t are functions of $(e_i, X_{e_i})_{i=1}^t$, as is the choice of e_{t+1}, the $(t+1)^{th}$ edge chosen by the FO-CRS. Thus, $\mathcal{M}_t$, F_t and e_{t+1} are measurable with respect to $\mathcal{H}_t$, the sigma-algebra generated from $(e_i, X_{e_i})_{i=1}^t$ (here $\mathcal{H}_0 := \{\emptyset, \Omega\}$, the trivial sigma-algebra). We refer to $\mathcal{H}_t$ as the *history* after t steps. It will also be convenient to define the *residual graph* R_t after t steps by

$$R_t = G[F_t] \setminus \{e_1, \ldots, e_t\}. \tag{6.1}$$

Notice that R_t contains precisely those edges of G which can feasibly added to the matching $\mathcal{M}_t$ after $e_1, \ldots, e_t$ have been processed.

Ideally, we would next upper bound $\mathbb{E}[|\mathcal{M}_{t+1}| - |\mathcal{M}_t| \mid \mathcal{H}_t]$ for each $t = 0, \ldots, n^2 - 1$ in terms of t and $|\mathcal{M}_t|$, and apply the one-sided differential equation method of Bennett and MacRury (2025) to conclude that $|\mathcal{M}_t|/n$ is dominated by the solution of a differential equation. This is the approach taken in MacRury et al. (2025) for RCRS, where the authors use that since e_{t+1} is distributed uniformly at random amongst $G \setminus \{e_1, \ldots, e_t\}$, $\mathbb{E}[|\mathcal{M}_{t+1}| - |\mathcal{M}_t| \mid \mathcal{H}_t]$ is upper bounded by roughly $\frac{1}{n}\left(1 - \frac{|\mathcal{M}_t|}{n}\right)^2$, and so $|\mathcal{M}_{n^2}|/n$ is at most $w(1) = 1/2$, where w is the unique solution to the differential equation $w'(\tau) = (1 - w(\tau))^2$ with $w(0) = 0$.

However, since e_{t+1} is chosen adaptively by the FO-CRS, there is no such simple upper bound. Instead, our approach is to define a *surplus function* $\mathcal{S}$, which assigns a non-negative value $\mathcal{S}(R_t) \geq 0$ to the residual subgraph R_t. Roughly speaking, $\mathcal{S}(R_t)$ approximately upper bounds the expected matching size available to an FO-CRS which begins with the subgraph R_t. Given such a function $\mathcal{S}$, we consider the random variable

$$Q_t := |\mathcal{M}_t| + \mathcal{S}(R_t), \tag{6.2}$$

and argue that for an appropriately chosen $\mathcal{S}$, $(Q_t)_{t=0}^{n^2}$ forms a supermartingale for which $Q_0 = \mathcal{S}(R_0) = \mathcal{S}(G)$ is our target upper bound of $(1+o(1))\left(1-\ln\left(2-\frac{1}{e}\right)\right)n$. Assuming such a choice of $\mathcal{S}$ is possible, Lemma 6.1 follows easily, as we show in Lemma 6.2.

In order to get a supermartingale, we need to ensure that when e_{t+1} is processed,

$$\mathbb{E}[Q_{t+1} - Q_t \mid \mathcal{H}_t] \leq 0. \tag{6.3}$$

There are two main cases we must consider for (6.3) to hold. If e_{t+1} is rejected by the FO-CRS, then $|\mathcal{M}_{t+1}| = |\mathcal{M}_t|$, and yet $R_{t+1} = R_t \setminus e_{t+1}$. Thus, we require $\mathcal{S}(R_t \setminus e_{t+1}) \leq \mathcal{S}(R_t)$. On the other hand, suppose the FO-CRS accepts e_{t+1} when it is active. Then, if $X_{e_{t+1}} = 1$, we have that $R_{t+1} = R_t - u_{t+1} - v_{t+1}$ and $|\mathcal{M}_{t+1}| = |\mathcal{M}_t| + 1$. Alternatively, if $X_{e_{t+1}} = 0$, then $R_{t+1} = R_t \setminus e_{t+1}$ and $|\mathcal{M}_{t+1}| = |\mathcal{M}_t|$. By averaging over $X_{e_{t+1}} \in \{0,1\}$, this leads to the inequality $(1-p)\mathcal{S}(R_t \setminus e_{t+1}) + p\big(1 + \mathcal{S}(R_t - u_{t+1} - v_{t+1})\big) \leq \mathcal{S}(R_t)$.

Given this motivation, we formalize the properties of a surplus function in the below definition:

Definition 3. The domain of a *surplus function* $\mathcal{S}$ is the set of all subgraphs $H \subseteq G$, where $\mathcal{S}(H) \geq 0$ denotes the non-negative value assigned to $H \subseteq G$. Moreover, for each $H \subseteq G$ and $(u,v) \in E(H)$,

$$\mathcal{S}(H \setminus (u,v)) \leq \mathcal{S}(H) \tag{6.4}$$

$$(1-p)\mathcal{S}(H \setminus (u,v)) + p\big(1 + \mathcal{S}(H - u - v)\big) \leq \mathcal{S}(H). \tag{6.5}$$

Finally,

$$\mathcal{S}(G) \leq (1+o(1))\left(1 - \ln\left(2 - \frac{1}{e}\right)\right)n. \tag{6.6}$$

Lemma 6.2 (proven in §D.1)**.** *Suppose that $\mathcal{S}$ is a surplus function. Then, $(Q_t)_{t=1}^{n^2}$ as defined in (6.2) is a supermartingale with respect to the filtration $(\mathcal{F}_t)_{t=1}^{n^2}$. Moreover, $\mathbb{E}[|\mathcal{M}_{n^2}|] \leq (1 + o(1))\left(1 - \ln\left(2 - \frac{1}{e}\right)\right)n$.*

6.1 Designing a Surplus Function

We now construct a surplus function satisfying Definition 3. Since we are interested in the limit as $n \to \infty$, we may assume that $n \geq 2$, and hence that $p = 1/n \in (0,1)$. Write $q := 1 - p$. Let $(B_v)_{v \in V(G)}$ be mutually independent random variables with $B_v \sim \mathrm{Ber}(p)$. We say that $v \in V$ is *marked*, provided $B_v = 1$. For each subgraph $H \subseteq G$, define $\mathcal{I}_H$ to be the event that the marked vertices form an independent set in H. Formally,

$$\mathcal{I}_H := \bigcap_{(u,v) \in E(H)} \{B_u B_v = 0\}. \tag{6.7}$$

Since $p < 1$, it is clear that $0 < \Pr[\mathcal{I}_H] \leq 1$. We next define the following function whose input in a subgraph $H \subseteq G$:

$$\mathcal{S}(H) := -n \ln \Pr[\mathcal{I}_H]. \tag{6.8}$$

Lemma 6.3. *The function $\mathcal{S}$ defined in (6.8) is a surplus function in the sense of Definition 3.*

In order to motivate $\mathcal{S}$, it is helpful to first see that it attains the desired value on G as in Definition 3. Recall that G is the complete bipartite graph on $2n$ vertices. A marked set is independent in G if and only if at least one of the two parts U_1, U_2 contains no marked vertex. Inclusion–exclusion therefore gives

$$\Pr[\mathcal{I}_G] = 2(1-p)^n - (1-p)^{2n}. \tag{6.9}$$

Since $p = 1/n$, it follows that

$$\begin{aligned}\mathcal{S}(G) &= -n \ln \left(2(1-1/n)^n - (1-1/n)^{2n}\right) \\ &= \left(1 - \ln\left(2 - \frac{1}{\mathrm{e}}\right) + o(1)\right) n. \end{aligned} \tag{6.10}$$

It remains to verify the two remaining conditions in Definition 3. We focus on (6.5), since it is harder to prove. Substituting (6.8) into that inequality, dividing by n, and using $p/n = p^2$ shows that it is equivalent to

$$\ln \Pr[\mathcal{I}_H] \le q \ln \Pr[\mathcal{I}_{H\setminus\{e\}}] + p \ln \Pr[\mathcal{I}_{H-u-v}] - p^2.$$

Exponentiating, this is

$$\Pr[\mathcal{I}_H] \le \mathrm{e}^{-p^2} \Pr[\mathcal{I}_{H\setminus\{e\}}]^q \Pr[\mathcal{I}_{H-u-v}]^p. \tag{6.11}$$

The next lemma proves the slightly stronger inequality obtained by replacing e^{-p^2} with $1 - p^2$.

Lemma 6.4 (proven in §D.2)**.** *For every subgraph $H \subseteq G$ and every edge $e = (u, v) \in E(H)$,*

$$\Pr[\mathcal{I}_H] \le (1-p^2) \Pr[\mathcal{I}_{H\setminus\{e\}}]^q \Pr[\mathcal{I}_{H-u-v}]^p. \tag{6.12}$$

Equivalently,

$$\Pr[\mathcal{I}_H \mid \mathcal{I}_{H-u-v}] \le (1-p^2) \Pr[\mathcal{I}_{H\setminus\{e\}} \mid \mathcal{I}_{H-u-v}]^q. \tag{6.13}$$

We now are ready to complete the proof of Lemma 6.3.

Proof of Lemma 6.3. Since $\Pr[\mathcal{I}_H] \le 1$, the function $\mathcal{S}$ is non-negative. If $e = (u, v) \in E(H)$, then deleting e removes one of the constraints defining $\mathcal{I}_H$. Thus $\mathcal{I}_H \subseteq \mathcal{I}_{H\setminus\{e\}}$, and hence $\Pr[\mathcal{I}_{H\setminus\{e\}}] \ge \Pr[\mathcal{I}_H]$. Since $x \mapsto -\ln x$ is decreasing, it follows from (6.8) that $\mathcal{S}(H \setminus \{e\}) \le \mathcal{S}(H)$. This is (6.4). Next, by Lemma 6.4 and the elementary inequality $1 - p^2 \le \mathrm{e}^{-p^2}$, (6.11) holds. As observed above, this is equivalent to (6.5). Finally, (6.10) verifies (6.6). Thus all of the conditions in Definition 3 are satisfied. □

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A Deferred Proofs from Section 3

A.1 Proof of Lemma 3.4

Lemma 3.4 (proven in §A.1)**.** *When lower bounding* $\Pr[e \in \mathcal{M} \mid X_e = 1]$*, we may assume that the edges of* $T \cup \partial T$ *are processed "bottom up": The edges between* $D^i(e)$ *and* $D^{i+1}(e)$ *are processed before the edges between* $D^{i-1}(e)$ *and* $D^i(e)$ *for* $i = 1, 2, \ldots, \ell$*.*

We roughly follow the argument of Kulkarni et al. (2022). For the tree T, define the witness tree $W(T)$. An edge f of $T \cup \partial T$ is in $W(T)$ if it is observed before every edge f' that lies above it in T.[8] By definition, the edges of $W(T)$ are observed in a bottom-up fashion. Note that whether the Algorithm 1 matches e is a function of the edges in $W(T)$, and the decision it makes on those edges. Thus, as far as bounding $\Pr[e \in \mathcal{M} \mid X_e = 1]$ is concerned, we can assume that $T \cup \partial T = W(T)$.

A few notes about this identification: identifying $T \cup \partial T$ with $W(T)$ might replace some of the x_e values with 0. For instance, if a vertex in $W(T)$ is a leaf, we think about it as a leaf of $T \cup \partial T$ after the identification. This might reduce the values of some of the x_e, but this is safe, since all we ever assume about the x_e in our arguments is that $\sum_{e \in \partial(u)} x_e \leq 1$, and this constraint remains satisfied even after reducing some x_e.

Furthermore, if $W(T)$ does not have an edge that goes from $D^\ell(e)$ to $D^{\ell+1}(e)$, there is nothing to prove, since in this case, the neighborhood of e is effectively a tree, so $\Pr[e \in \mathcal{M} \mid X_e = 1] \geq \alpha$. We will therefore assume that T still has depth ℓ even after $T \cup \partial T$'s identification with $W(T)$. This establishes the lemma.

A.2 Proof of Lemma 3.5

Lemma 3.5 (proven in §A.2)**.** *For each* $0 \leq j \leq \ell - 1$*,* $\varepsilon_{j+1}^{\max} \leq \varepsilon_j^{\max}(1 - \alpha)$*. Thus,* $\varepsilon_\ell^{\max} \leq (1 - \alpha)^\ell$*, since* $\varepsilon_0^{\max} \leq 1$*.*

We first establish the following preliminary proposition:

Proposition A.1. *Fix* $k \in \mathbb{N}$*,* $\alpha \in (0, 1)$*, and* $\varepsilon, \bar{x} \in [0, 1]$*. Suppose* $\boldsymbol{x} = (x_i)_{i=1}^k$ *has non-negative entries, and* $\sum_{i=1}^k x_i = \bar{x}$*. Then, by maximizing (respectively, minimizing) over all such* $\boldsymbol{x}$*:*

$$\max_{\boldsymbol{x}} \prod_{i=1}^{k} \left(1 + \frac{\varepsilon \alpha x_i}{1 - \alpha \sum_{j \leq i} x_j}\right) = 1 + \frac{\varepsilon \alpha \bar{x}}{1 - \alpha \bar{x}}, \min_{\boldsymbol{x}} \prod_{i=1}^{k} \left(1 - \frac{\varepsilon \alpha x_i}{1 - \alpha \sum_{j \leq i} x_j}\right) = 1 - \frac{\varepsilon \alpha \bar{x}}{1 - \alpha \bar{x}}.$$

Proof. Let us concentrate first on establishing the maximizing version of the proposition. We prove

[8] $W(T)$ is not literally a tree because it might have edges between $D^\ell(e)$ and $D^{\ell+1}(e)$ which create cycles. Here, we follow the notation in Kulkarni et al. (2022).

the statement by induction on k. First, let us show that the statement is true for $k = 2$. Note that:

$$\begin{aligned}
& \left(1 + \frac{\varepsilon\alpha x_1}{1-\alpha x_1}\right) \cdot \left(1 + \frac{\varepsilon\alpha x_2}{1-\alpha(x_1+x_2)}\right) \leq 1 + \frac{\varepsilon\alpha(x_1+x_2)}{1-\alpha(x_1+x_2)} \\
\iff \quad & 1 + \frac{\varepsilon\alpha x_1}{1-\alpha x_1} + \frac{\varepsilon\alpha x_2}{1-\alpha(x_1+x_2)} + \frac{\varepsilon^2\alpha^2 x_1 x_2}{(1-\alpha x_1)(1-\alpha(x_1+x_2))} \leq 1 + \frac{\varepsilon\alpha(x_1+x_2)}{1-\alpha(x_1+x_2)} \\
\iff \quad & \frac{\varepsilon\alpha x_1}{1-\alpha x_1} + \frac{\varepsilon^2\alpha^2 x_1 x_2}{(1-\alpha x_1)(1-\alpha(x_1+x_2))} \leq \frac{\varepsilon\alpha x_1}{1-\alpha(x_1+x_2)} \\
\iff \quad & \varepsilon\alpha x_1(1-\alpha(x_1+x_2)) + \varepsilon^2\alpha^2 x_1 x_2 \leq \varepsilon\alpha x_1(1-\alpha x_1) \\
\iff \quad & -\varepsilon\alpha^2 x_1 x_2 + \varepsilon^2\alpha^2 x_1 x_2 \leq 0 \\
\iff \quad & \varepsilon\alpha^2 x_1 x_2(\varepsilon - 1) \leq 0.
\end{aligned}$$

It follows immediately that the inequality we started with is true, and furthermore, that the inequality is in fact an equality when x_1 or x_2 is equal to 0. Hence the statement is true for $k = 2$, and the base case is established.

Suppose we have established the statement for $k = r$, and we are attempting to establish the statement for $k = r + 1$. Then, notice that:

$$\begin{aligned}
\prod_{i=1}^{r+1}\left(1 + \frac{\varepsilon\alpha x_i}{1-\alpha\sum_{j\leq i} x_j}\right) &\leq \left(1 + \frac{\varepsilon\alpha(x_1+x_2)}{1-\alpha(x_1+x_2)}\right)\prod_{i=3}^{r+1}\left(1 + \frac{\varepsilon\alpha x_i}{1-\alpha\sum_{j\leq i} x_j}\right) \\
&\leq 1 + \frac{\varepsilon\alpha\bar{x}}{1-\alpha\bar{x}}.
\end{aligned}$$

The first inequality follows from an application of the case $k = 2$, and the second follows from the inductive hypothesis. Furthermore, the inequality is tight when $x_2 = x_3 = \ldots = x_n = 0$. The desired result follows.

The proof of the minimizing version of the proposition is almost entirely identical, but we include a proof for completness. We prove the statement by induction on k. First, let us show that the statement is true for $k = 2$. Note that:

$$\begin{aligned}
& \left(1 - \frac{\varepsilon\alpha x_1}{1-\alpha x_1}\right) \cdot \left(1 - \frac{\varepsilon\alpha x_2}{1-\alpha(x_1+x_2)}\right) \geq 1 - \frac{\varepsilon\alpha(x_1+x_2)}{1-\alpha(x_1+x_2)} \\
\iff \quad & 1 - \frac{\varepsilon\alpha x_1}{1-\alpha x_1} - \frac{\varepsilon\alpha x_2}{1-\alpha(x_1+x_2)} + \frac{\varepsilon^2\alpha^2 x_1 x_2}{(1-\alpha x_1)(1-\alpha(x_1+x_2))} \geq 1 - \frac{\varepsilon\alpha(x_1+x_2)}{1-\alpha(x_1+x_2)} \\
\iff \quad & -\frac{\varepsilon\alpha x_1}{1-\alpha x_1} + \frac{\varepsilon^2\alpha^2 x_1 x_2}{(1-\alpha x_1)(1-\alpha(x_1+x_2))} \geq -\frac{\varepsilon\alpha x_1}{1-\alpha(x_1+x_2)} \\
\iff \quad & -\varepsilon\alpha x_1(1-\alpha(x_1+x_2)) + \varepsilon^2\alpha^2 x_1 x_2 \geq -\varepsilon\alpha x_1(1-\alpha x_1) \\
\iff \quad & \varepsilon\alpha^2 x_1 x_2 + \varepsilon^2\alpha^2 x_1 x_2 \geq 0.
\end{aligned}$$

It follows immediately that the inequality we started with is true, and furthermore, that the inequality is in fact an equality when x_1 or x_2 is equal to 0. Hence the statement is true for $k = 2$, and the base case is established.

Suppose we have established the statement for $k = r$, and we are attempting to establish the statement for $k = r + 1$. Then, notice that:

$$\prod_{i=1}^{r+1}\left(1-\frac{\varepsilon\alpha x_i}{1-\alpha\sum_{j\le i}x_j}\right)\ge\left(1-\frac{\varepsilon\alpha(x_1+x_2)}{1-\alpha(x_1+x_2)}\right)\prod_{i=3}^{r+1}\left(1-\frac{\varepsilon\alpha x_i}{1-\alpha\sum_{j\le i}x_j}\right)$$
$$\ge 1-\frac{\varepsilon\alpha\bar{x}}{1-\alpha\bar{x}}.$$

The first inequality follows from an application of the case $k=2$, and the second follows from the inductive hypothesis. Furthermore, the inequality is tight when $x_2=x_3=\ldots=x_n=0$. The desired result follows. □

We are ready to return to the proof of Lemma 3.5:

Proof of Lemma 3.5. Fix $r\in D^{\ell-j-1}(e)$, and assume that r has $k\ge 1$ children, as otherwise $\varepsilon_r=0$. Then, since $|\varepsilon_{r_i}|\le\varepsilon_j^{\max}$ for $i\in[k]$,

$$\begin{aligned}\varepsilon_r &= 1-\prod_{i=1}^{k}\left(1+\frac{\varepsilon_{r_i}\alpha x_{e_i}}{1-\alpha\sum_{i'\le i}x_{r,r_{i'}}}\right)\\ &\le 1-\prod_{i=1}^{k}\left(1-\frac{\varepsilon_j^{\max}\alpha x_{e_i}}{1-\alpha\sum_{i'\le i}x_{r,r_{i'}}}\right)\le 1-\left(1-\frac{\varepsilon_j^{\max}\alpha x_r}{1-\alpha x_r}\right)\le\varepsilon_j^{\max}(1-\alpha),\end{aligned}$$

where the second inequality uses Proposition A.1, and the last uses $x_r\le 1$. Similarly, $\varepsilon_r\ge -\varepsilon_j^{\max}(1-\alpha)$, and so $|\varepsilon_r|\le\varepsilon_j^{\max}\cdot(1-\alpha)$ for all $r\in D^{\ell-j-1}(e)$. □

A.3 Proof of Proposition 3.6

Proposition 3.6 (proven in §A.3)**.** *Suppose $x,y,z\in(0,1)$ satisfy $x\ge yz$. There exists a coupling of Bernoulli random variables $X\sim\mathrm{Ber}(x)$, $Z\sim\mathrm{Ber}(z)$, and $Y\sim\mathrm{Ber}(y)$, where Y,Z are independent, and $X\ge Y\cdot Z$.*

We first handle the case when $x=yz$. In this case, clearly $y,z\ge x$.

If $X=1$, then set $Y=Z=1$. Otherwise, if $X=0$, there are three outcomes to assign probability mass to: $Y=1$ and $Z=0$, $Y=0$ and $Z=1$, and $Y=0$ and $Z=0$. Denote the amount assigned to each by $a_{1,0},a_{0,1}$ and $a_{0,0}$, respectively. We wish to determine $a_{i,j}\in[0,1]$ for which the following hold:

1. $a_{0,0}+a_{0,1}+a_{1,0}=1$
2. $(1-x)a_{1,0}+x=y$
3. $(1-x)a_{0,1}+x=z$

Observe that $a_{1,0}=(y-x)/(1-x)$ and $a_{0,1}=(z-x)/(1-x)$. Clearly each is within $[0,1]$, as $y-x,z-x\ge 0$. Moreover, $a_{0,0}+a_{0,1}+a_{1,0}=(y+z-2x)/(1-x)+a_{0,0}$. So it suffices to show that $(y+z-2x)/(1-x)\le 1$ to conclude $a_{0,0}\in[0,1]$. But this is equivalent to $y+z\le 1+x=1+yz$. Since $y+z\le 1+yz$ for any $y,z\in[0,1]$, $a_{0,0}=1-a_{0,1}-a_{1,0}\in[0,1]$.

Finally, observe that Y and Z are independent. This is because $\Pr[Y=1,Z=1]=\Pr[X=1]=x=yz$.

Let us now suppose that $x\ge yz$. We first couple $X'\sim\mathrm{Ber}(yz)$ and such that $X\ge X'$. By applying the previous construction to X', we get Y and Z which satisfy $X\ge X'=YZ$.

A.4 Proof of Lemma 3.8

Lemma 3.8 (proven in §A.4)**.** *Conditional on G' (equivalently $(Y_e)_{e\in E}$), the random variables $(Z'_e)_{e\in E}$ are independently distributed with marginals described by $(z'_e)_{e\in E}$.*

Observe that the fractional matching $(z'_e)_{e\in E}$ is a function of $(Y_e)_{e\in E}$ (i.e., it is $(Y_e)_{e\in E}$-measurable). Thus, if we define the random subset $\mathcal{S} := \{e \in E : z'_e > 0\}$, then $\mathcal{S}$ is also a function of $(Y_e)_{e\in E}$. On the other hand, observe that once $\mathcal{S}$ is fixed, $(Z'_e)_{e\in\mathcal{S}}$ and $(Z_e)_{e\in\mathcal{S}}$ are equal, and so distributed identically. Since the random variables $((Y_e)_{e\in E}, (Z_e)_{e\in E})$ are independent by construction, and $Z'_e = 0$ for all $e \in E \setminus \mathcal{S}$, the claim follows.

A.5 Proof of Lemma 3.9

Lemma 3.9 (proven in §A.5)**.** *For each $e \in E$,* $\Pr[z'_e = 1/\log(1/\varepsilon) \mid Y_e = 1] \geq 1 - 2\exp\left(-\Omega(\log^{1/2}(1/\varepsilon))\right)$.

Consider the randomly generated fractional matching $(z'_f)_{f\in E}$ output by Algorithm 2. Given $e = (u, v) \in E$, define $z'_v(e) := \sum_{\substack{f\in\partial(v):\\ f<e}} z'_f$ and $z'_u(e) := \sum_{\substack{f\in\partial(u):\\ f<e}} z'_f$. Observe that $z'_e = 1/\log(1/\varepsilon)$ if and only if $Y_e = 1$ and $\max\{z'_v(e), z'_u(e)\} \leq 1 - \frac{1}{\log(1/\varepsilon)}$. On the other hand, since $z'_v(e)$ and $z'_u(e)$ are independent of Y_e, to prove the lemma it suffices to show that

$$\Pr\left[\max\{z'_v(e), z'_u(e)\} \leq 1 - \frac{1}{\log(1/\varepsilon)}\right] \geq 1 - 2\exp\left(-\Omega(\log^{1/2}(1/\varepsilon))\right) \tag{A.1}$$

We begin by considering $z'_v(e)$. Observe that

$$z'_v(e) = \sum_{\substack{f\in\partial(v):\\ f<e}} z'_f \leq \sum_{f\in\partial(v)\setminus e} \frac{Y_f}{\log(1/\varepsilon)}, \tag{A.2}$$

where $(Y_f)_{f\in\partial(v)\setminus e}$ are independently drawn Bernoulli random-variables. Moreover,

$$\sum_{f\in\partial(v)\setminus e} \mathbb{E}[Y_f] = \sum_{f\in\partial(v)\setminus e} (1 - \log^{-1/4}(\varepsilon^{-1}))x_f \log(1/\varepsilon) \leq (1 - \log^{-1/4}(\varepsilon^{-1}))\log(1/\varepsilon),$$

since $\sum_{f\in\partial(v)} x_f \leq 1$. Setting $\delta = \log^{-1/4}(\varepsilon^{-1})$ and $\mu = (1 - \log^{-1/4}(\varepsilon^{-1}))\log(1/\varepsilon)$, we can thus apply Chernoff bound to get that for n sufficiently large,

$$\Pr\left[\sum_{f\in\partial(v)\setminus e} Y_f \geq (1+\delta)\left(1 - \log^{-1/4}(\varepsilon^{-1})\right)\log(1/\varepsilon)\right] \leq \exp\left(-\frac{\delta^2\mu}{2+\delta}\right) \leq \exp\left(-\frac{\log^{1/2}(\varepsilon^{-1})}{3}\right).$$

On the other hand, $(1+\delta)\left(1 - \log^{-1/4}(\varepsilon^{-1})\right) \leq 1 - \log^{-1}(1/\varepsilon)$, so

$$\Pr\left[\sum_{f\in\partial(v)\setminus e} Y_f \geq \log(1/\varepsilon)\left(1 - \frac{1}{\log(1/\varepsilon)}\right)\right] \leq \exp\left(-\frac{\log^{1/2}(\varepsilon^{-1})}{3}\right).$$

Thus, $\Pr[z_v(e) > 1 - 1/\log(1/\varepsilon)] \leq \exp\left(-\Omega(\log^{1/2}(1/\varepsilon))\right)$ after applying this bound to (A.2). An analogous argument applies to $z'_u(e)$, and so (A.1) follows after applying a union bound. The proof of the lemma is thus complete.

A.6 Proof of Lemma 3.10

Lemma 3.10 (proven in §A.6)**.** *For each $e \in E$,* $\Pr[C_\ell(e) \mid Y_e = 1] = O(\varepsilon(\log(1/\varepsilon))^{2\ell+1})$.

In order to prove the lemma, we first upper bound the probability there exists a cycle in $N^\ell_{G'}(e)$ which contains e. Afterwards, we upper bound the probability there exists a cycle in $N^\ell_{G'}(e)$ which does *not* contain e. Both upper bounds are of the form $O(\varepsilon(\log(1/\varepsilon))^{2\ell+1})$, so a simple union bound will then complete the proof.

Let N_k be a random variable counting the number of k length cycles in G' which contain e. Since we only care about cycles of this form in the ℓ-neighborhood of e, it suffices to prove that

$$\Pr\left[\sum_{k=3}^{2\ell+2} N_k > 0 \middle| Y_e = 1\right] = O\left(\varepsilon(\log(1/\varepsilon))^{2\ell+1}\right). \tag{A.3}$$

Assume for now that

$$\mathbb{E}[N_k \mid Y_e = 1] \leq \varepsilon(\log(1/\varepsilon))^{k-1}. \tag{A.4}$$

In this case, by Markov's inequality and linearity of expectation,

$$\Pr\left[\sum_{k=3}^{2\ell+2} N_k > 0 \middle| Y_e = 1\right] \leq \sum_{k=3}^{2\ell+2} \varepsilon(\log(1/\varepsilon))^{k-1} = O\left(\varepsilon(\log(1/\varepsilon))^{2\ell+1}\right),$$

where the final inequality holds from the geometric series formula.

In order to establish (A.4), let $w_1, w_2, \ldots, w_{k-2}$ denote $k-2$ distinct vertices in G, all of them different from u and v. For convenience, let us write $u = w_0$, and $v = w_{k-1}$. Then, we have that:

$$\mathbb{E}[N_k \mid Y_e = 1] = \sum_{w_1,w_2,\ldots,w_{k-2}} \prod_{i=0}^{k-2} y_{w_i w_{i+1}} \leq (\log(1/\varepsilon))^{k-1} \sum_{w_1,w_2,\ldots,w_{k-2}} \prod_{i=0}^{k-2} x_{w_i w_{i+1}},$$

since $y_e \leq x_e \log(1/\varepsilon)$. Now, we know $x_{w_{k-2}v} \leq \epsilon$, for all choices of w_{k-2}. Therefore, it follows that:

$$\mathbb{E}[N_k \mid Y_e = 1] \leq \varepsilon(\log(1/\varepsilon))^{k-1} \sum_{w_1,w_2,\ldots,w_{k-2}} \prod_{i=0}^{k-3} x_{w_i w_{i+1}}.$$

Finally, we will show using a simple induction on k that

$$\sum_{w_1,w_2,\ldots,w_{k-2}} \prod_{i=0}^{k-3} x_{w_i w_{i+1}} \leq 1. \tag{A.5}$$

The statement is true for $k = 3$, since $\sum_{w_1} x_{uw_1} \leq 1$ as $\boldsymbol{x}$ is fractional matching. Suppose we have already established the statement for $k = r \geq 3$. Then, for $k = r+1$,

$$\begin{aligned}\sum_{w_1,w_2,\ldots,w_{r-1}} \prod_{i=0}^{r-2} x_{w_i w_{i+1}} &= \sum_{w_1,w_2,\ldots,w_{r-2}} \prod_{i=0}^{r-3} x_{w_i w_{i+1}} \cdot \left(\sum_{w_{r-1}} x_{w_{r-2} w_{r-1}}\right) \\ &\leq \sum_{w_1,w_2,\ldots,w_{r-2}} \prod_{i=0}^{r-3} x_{w_i w_{i+1}} \leq 1,\end{aligned}$$

where the first inequality follows from the fact that $\boldsymbol{x}$ is a fractional matching, and the second from the induction hypothesis. We conclude that (A.5) as desired, and so (A.4) holds.

Next, let us consider cycles in $N^{\ell}_{G'}(e)$ which do *not* contain the edge e. The existence of such a cycle guarantees the existence of a path of length $k \leq \ell - 1$ from either u of v to a distinguished vertex w (possibly equal to u or v), and a cycle of length at most $2(\ell - k) + 1$ that contains w. Note that we can upper bound the probability that such a structure by:

$$\sum_{w \in V, k \leq \ell - 1} \Pr[\exists \text{ a path of length } k \text{ from } u \text{ to } w] \Pr[\exists \text{ a cycle of length} \leq 2(\ell - k) + 1 \text{ containing } w].$$

Now, first of all,

$$\Pr[\text{there is a cycle of length} \leq 2(\ell - k) + 1 \text{ containing } w] = O(\varepsilon(\log(1/\varepsilon))^{2(\ell - k) + 1}),$$

by an argument analogous to the one used to establish (A.3) (we let $w_0 = w_{\ell - k - 1} = w$). Furthermore, note that if we write $u = w_0$, and $w = w_k$, then:

$$\begin{aligned}
\sum_{w \in G} \Pr[\text{there is path of length } k \text{ from } u \text{ or } v \text{ to } w] &\leq 2 \sum_{w \in G} \Pr[\text{there is path of length } k \text{ from } u \text{ to } w] \\
&\leq 2 \sum_{w_1, w_2, \ldots, w_{k-1}, w_k} \prod_{i=0}^{k-1} y_{w_i w_{i+1}} \\
&\leq 2(\log(1/\varepsilon))^k \sum_{w_1, w_2, \ldots, w_{k-1}, w_k} \prod_{i=0}^{k-1} x_{w_i w_{i+1}} \\
&\leq 2(\log(1/\varepsilon))^k.
\end{aligned}$$

where the final inequality follows from (A.5). Finally it follows that the probability with which such a structure exists is at most:

$$\sum_{k=0}^{\ell - 1} O(\varepsilon(\log(1/\varepsilon))^{2(\ell - k) + 1}) \cdot 2(\log(1/\varepsilon))^k = O\left(\varepsilon(\log(1/\varepsilon))^{2\ell + 1}\right).$$

Together with the previous upper bound of (A.3), this completes the proof.

B Deferred Proofs from Section 4

B.1 Proof of Lemma 4.2

Lemma 4.2 (Small covariance regime: proven in §B.1)**.** $\theta \geq (1 + o(1))(c - (1 - c)^2)$.

Let suppose that $v_i \neq v_j$ satisfy $\theta = \mathrm{Cov}[A_{v_i}, A_{v_j}]$. Consider when (v_i, v_j) is presented to the OCRS in the second phase, and (v_i, v_j) is active. Observe that in order for (v_i, v_j) to be matched, both v_i and v_j must be *available* (i.e., $A_{v_i} = A_{v_j} = 1$). Thus, since A_{v_i} and A_{v_j} are independent of X_{v_i, v_j},

$$\begin{aligned}
c = \Pr[(v_i, v_j) \in \mathcal{M} \mid X_{v_i, v_j} = 1] \leq \Pr[A_{v_i} = 1, A_{v_j} = 1] &= \mathbb{E}[A_{v_i}]\mathbb{E}[A_{v_j}] + \mathrm{Cov}[A_{v_i}, A_{v_j}] \\
&\leq \left(1 - c\left(1 - \frac{2}{n}\right)\right)^2 + \theta,
\end{aligned}$$

where the final inequality uses (4.2) together with $A_{v_i} = 1 - M_{v_i}$, as well as the definition of θ. As such, after rearranging for θ and taking $n \to \infty$,

$$\theta \geq c - \left(1 - c\left(1 - \frac{2}{n}\right)\right)^2 = (1 + o(1))(c - (1 - c)^2).$$

B.2 Proof of Lemma 4.3

Lemma 4.3 (Large covariance regime: proven in §B.2)**.** $2c + c\exp(c + \theta/c - 1) \leq 1 + o(1)$.

In order to prove the lemma, we analyze the third and final phase. Let us assume that the edges of $\partial(u)$ are presented in order $(u, v_1), \ldots, (u, v_n)$ to the OCRS, and consider the arrival of the last edge $e_n = (u, v_n)$. Define $M_u(e_n)$ to be the indicator random variable for the event u is matched when e_n arrives. Observe then that conditional on $X_{e_n} = 1$, e_n is matched only if $M_{v_n} = 0$ *and* $M_u(e_n) = 0$. Thus, since X_{e_n} is independent of the latter two events,

$$\begin{aligned} c = \Pr[e_n \in \mathcal{M} \mid X_{e_n} = 1] &\leq \Pr[M_{v_n} = 0, M_u(e_n) = 0] \\ &= 1 - \Pr[M_{v_n} = 1] - \Pr[M_u(e_n) = 1] + \Pr[M_{v_n} = 1, M_u(e_n) = 1] \\ &= 1 - \Pr[M_{v_n} = 1](1 - \Pr[M_u(e_n) = 1 \mid M_{v_n} = 1]) - \Pr[M_u(e_n) = 1] \end{aligned}$$

Now, $\Pr[M_u(e_n) = 1] = \sum_{i=1}^{n-1} \Pr[(u, e_i) \in \mathcal{M}] = c\left(1 - \frac{1}{n}\right)$, so after simplifying and taking $n \to \infty$, we get that

$$2c + \Pr[M_{v_n} = 1](1 - \Pr[M_u(e_n) = 1 \mid M_{v_n} = 1]) \leq 1 + o(1). \tag{B.1}$$

We shall now prove that

$$\Pr[M_u(e_n) = 1 \mid M_{v_n} = 1] \leq (1 + o(1))(1 - \exp(c + \theta/\Pr[M_{v_n} = 1] - 1)). \tag{B.2}$$

This will complete the proof. To see this, first observe that by applying (B.2) to (B.1),

$$2c + \Pr[M_{v_n} = 1](\exp(c + \theta/\Pr[M_{v_n} = 1] - 1)) \leq 1 + o(1).$$

Thus, since $\Pr[M_{v_n} = 1] = c(1 - 2/n)$ by (4.2), $2c + c(\exp(c + \theta/c - 1)) \leq 1 + o(1)$ as required.

In order to prove (B.2), recall that $A_{v_n} = 1 - M_{v_n}$. We shall instead condition on $A_{v_n} = 0$, and upper bound $\Pr[M_u(e_n) = 1 \mid A_{v_n} = 0]$. Consider the vertices amongst $v_1, \ldots, v_{n-1}$ which were available after the first phase. Observe that if we condition on $A_{v_n} = 0$, then in order for $M_u(e_n) = 1$, there must be an active edge from u to one of these vertices. Thus,

$$\Pr[M_u(e_n) = 1 \mid A_{v_n} = 0] \leq \Pr[\cup_{j=1}^{n-1}\{X_{u,v_j} = 1\} \cap \{A_{v_j} = 1\} \mid A_{v_n} = 0]. \tag{B.3}$$

On the other hand, the edge states $(X_{u,v_j})_{j=1}^n$ are independent from $(A_{v_i})_{i=1}^n$, and so

$$\begin{aligned} \Pr[\cup_{j=1}^{n-1}\{X_{u,v_j} = 1\} \cap \{A_{v_j} = 1\} \mid (A_{v_i})_{i=1}^{n-1}, A_{v_n} = 0] &= 1 - \prod_{j=1}^{n-1}\left(1 - \frac{A_{v_j}}{n}\right) \\ &= (1 + o(1))\left(1 - \exp\left(-\frac{\sum_{j=1}^{n-1} A_{v_j}}{n}\right)\right). \end{aligned}$$

Now, the function $z \to 1 - \exp(-z)$ is concave for $z \in [0, 1]$. Thus, if we take expectations over the $(A_{v_i})_{i=1}^{n-1}$ random variables (while conditioning on $A_{v_n} = 0$) then Jensen's inequality implies that

$$\Pr[\cup_{j=1}^{n-1}\{X_{u,v_j} = 1\} \cap \{A_{v_j} = 1\} \mid A_{v_n} = 0] \leq (1 + o(1))\left(1 - \exp\left(-\frac{\sum_{j=1}^{n-1} \mathbb{E}[A_{v_j} \mid A_{v_n} = 0]}{n}\right)\right),$$

and so combined with (B.3), we get that

$$\Pr[M_u(e_n) = 1 \mid A_{v_n} = 0] \leq (1 + o(1))\left(1 - \exp\left(-\frac{\sum_{i=1}^{n-1} \mathbb{E}[A_{v_i} \mid A_{v_n} = 0]}{n}\right)\right). \tag{B.4}$$

As such, we focus on upper bounding $\sum_{i=1}^{n-1} \mathbb{E}[A_{v_i} \mid A_{v_n} = 0]$. We argue that for each $1 \le i \le n-1$, the events $A_{v_n} = 0$ and $A_{v_i} = 1$ are *negatively* correlated, which we quantify in terms of θ. Specifically, by the definition of θ, $\mathbb{E}[A_{v_i} \mid A_{v_n} = 1] \Pr[A_{v_n} = 1] = \mathbb{E}[A_{v_i}]\mathbb{E}[A_{v_n}] + \mathrm{Cov}[A_{v_i}, A_{v_n}] \ge \mathbb{E}[A_{v_i}]\mathbb{E}[A_{v_n}] + \theta$. Thus,

$$\begin{aligned}\mathbb{E}[A_{v_i} \mid A_{v_n} = 0] \Pr[A_{v_n} = 0] &= \mathbb{E}[A_{v_i}] - \mathbb{E}[A_{v_i} \mid A_{v_n} = 1] \Pr[A_{v_n} = 1] \\ &\le \mathbb{E}[A_{v_i}] - \mathbb{E}[A_{v_i}]\mathbb{E}[A_{v_n}] - \theta \\ &= \mathbb{E}[A_{v_i}] \Pr[A_{v_n} = 0] - \theta.\end{aligned}$$

Since $\Pr[A_{v_n} = 0] > 0$ for $n > 2$, and $\mathbb{E}[A_{v_i}] = 1 - c\left(1 - \frac{2}{n}\right)$, it follows that

$$\sum_{i=1}^{n-1} \mathbb{E}[A_{v_i} \mid A_{v_n} = 0] \le (1 + o(1))n \left((1 - c) - \frac{\theta}{\Pr[A_{v_n} = 0]}\right). \tag{B.5}$$

By applying (B.5) to (B.4), $\Pr[M_u(e_n) = 1 \mid A_{v_n} = 0] \le (1+o(1))(1-\exp(-1+c+\theta/\Pr[A_{v_n} = 0]))$, and so (B.2) is proven, thereby completing the proof.

B.3 Proof of Lemma 4.4

Lemma 4.4 (proven in §B.3)**.** *Suppose $n \to \infty$. Let $A_1, A_2, \ldots, A_n$ be random variables, with $\mathrm{Var}(A_i) < \delta$. Define $A = \sum_{i=1}^n A_i$, and $A_S = \sum_{i \in S} A_i$. Suppose $Var(A) \le \theta_n n^2$ and $\theta_n \ge \frac{1}{\sqrt{n}}$. Then, we can find $S \subseteq \{1, 2, \ldots, n\}$ with $|S| = m = \theta_n^{1/4} n$ such that $Var(A_S) \le 3\theta_n m^2$ for sufficiently large n.*

Since

$$\mathrm{Var}(A) = \sum_{i \ne j} \mathrm{Cov}(A_i, A_j) + \sum_{i=1}^n \mathrm{Var}(A_i),$$

it follows that $\sum_{i \ne j} \mathrm{Cov}(A_i, A_j) \le \theta_n n^2$. On the other hand,

$$\begin{aligned}\sum_{|S|=m} \mathrm{Var}(A_S) &\le \sum_{|S|=m} \left(\sum_{i \ne j, i,j \in S} \mathrm{Cov}(A_i, A_j) + \delta m \right) \\ &= \left(\sum_{|S|=m} \sum_{i \ne j, i,j \in S} \mathrm{Cov}(A_i, A_j) \right) + \binom{n}{m} \cdot \delta m \\ &= \sum_{i \ne j} \binom{n-2}{m-2} \mathrm{Cov}(A_i, A_j) + \binom{n}{m} \cdot \delta m.\end{aligned}$$

It follows that

$$\begin{aligned}\frac{\sum_{|S|=m} \mathrm{Var}(A_S)}{\binom{n}{m}} &\le \frac{\binom{n-2}{m-2}}{\binom{n}{m}} \left(\sum_{i \ne j} \mathrm{Cov}(A_i, A_j) \right) + \delta m \\ &\le \frac{m(m-1)}{n(n-1)} \theta_n n^2 + \delta m \\ &\le 2\theta_n m^2 + \delta m.\end{aligned}$$

Next, notice that,

$$\begin{aligned}\delta m \le \theta_n m^2 &\iff \delta \le \theta_n m \\ &\iff \delta \le \theta_n^{5/4} n \\ &\iff \delta \le \theta_n^{5/4} n \\ &\iff \frac{\delta^{4/5}}{n^{4/5}} \le \theta_n.\end{aligned}$$

Since we assumed that $\theta_n \ge \frac{1}{\sqrt{n}}$, the last inequality is true for sufficiently large n, so we conclude that:

$$\frac{\sum_{|S|=m} \mathrm{Var}(A_S)}{\binom{n}{m}} \le 3\theta_n m^2.$$

Since the average value of $\mathrm{Var}(A_S)$ is $\le 3\theta_n m^2$, it follows that there exists S such that $\mathrm{Var}(A_S) \le 3\theta_n m^2$, as required.

B.4 Proof of Lemma 4.6

Lemma 4.6 (proven in §B.4)**.** *Suppose* $\Pr[A_{end} > T] \le \varepsilon$*. Then,*

$$\Pr[\mathcal{B} \cap \mathcal{C}] \le \frac{T}{n} \Pr[\mathcal{C}] + \varepsilon \le \frac{T}{n}\left(1 - \frac{c(n-m)}{n}\right) + \varepsilon.$$

First, note that

$$\Pr[\mathcal{B} \cap \mathcal{C}] = \mathbb{E}\left[\mathbb{1}_{\mathcal{C}}\left(1 - \left(1 - \frac{1}{n}\right)^{A_{end}}\right)\right],$$

since $\mathcal{B}$ occurs exactly when $X_{u,v} = 1$ for some v with $A_{u,v} = 1$, and conditional on A_{end} and $\mathcal{C}$, the activity statuses of the last m edges are still independent, with probability $1/n$ of being active. Fixing a threshold T, with $\Pr[A_{end} > T] \le \varepsilon$, it follows that:

$$\begin{aligned}\mathbb{E}\left[\mathbb{1}_{\mathcal{C}}\left(1 - \left(1 - \frac{1}{n}\right)^{A_{end}}\right)\right] &\le \frac{T}{n}\Pr[\mathcal{C}] + \Pr[A_{end} > T] \\ &\le \frac{T}{n}\Pr[\mathcal{C}] + \varepsilon.\end{aligned}$$

The second inequality in the lemma follows from $\Pr[\mathcal{C}] \le 1 - c(n-m)/n$.

B.5 Proof of Theorem 4.7

Theorem 4.7 (proven in §B.5)**.** *Any OCRS is at most* $(\gamma + o(1))$*-selectable on* $(G, \boldsymbol{x})$ *where* γ *is the unique solution to the equation:* $0 = 1 - 2\gamma - \gamma\exp(\gamma - 1)$*, and* $\gamma \approx 0.3929$*.*

The proof in this setting will, broadly speaking, resemble the proof of Lemma 4.3, with some key differences. We will work with the same sequence of input graphs useful in the proof of Theorem 4.5, and in the first phase, the edges are presented in the same order. However, we will choose a different order for the edges in $\partial(u)$.

Once again, let us fix an OCRS for $(G, \boldsymbol{x})$ which returns a matching $\mathcal{M}$, and is c-selectable for $c \in (0,1)$, i.e., $\Pr[e \in \mathcal{M} \mid X_e = 1] = c$ for each edge e of G. We will assume that $\mathrm{Var}[\sum_{i=1}^n A_{v_i}] \geq \theta n^2$, for some $\theta \geq 0$. Before we explain the order in which the edges of $\partial(u)$ are presented, we note that since

$$\sum_{j=1}^{n} \left(\mathbb{E}[A_{v_j} \sum_{i=1}^{n} A_{v_i}] - \mathbb{E}[A_{v_j}]\mathbb{E}[\sum_{i=1}^{n} A_{v_i}] \right) = \mathrm{Var}[\sum_{i=1}^{n} A_{v_i}] \geq \theta n^2,$$

we can find j such that

$$\mathbb{E}[A_{v_j} \sum_{i=1}^{n} A_{v_i}] - \mathbb{E}[A_{v_j}]\mathbb{E}[\sum_{i=1}^{n} A_{v_i}] \geq \theta n.$$

But note that

$$\begin{aligned}
& \mathbb{E}[A_{v_j} \sum_{i=1}^{n} A_{v_i}] - \mathbb{E}[A_{v_j}]\mathbb{E}[\sum_{i=1}^{n} A_{v_i}] \geq \theta n \\
\iff \quad & \mathbb{E}[\sum_{i=1}^{n} A_{v_i} \mid A_{v_j} = 1] \Pr[A_{v_j} = 1] - (1-c)^2 n \geq \theta n \\
\iff \quad & \mathbb{E}[\sum_{i=1}^{n} A_{v_i}] - \mathbb{E}[\sum_{i=1}^{n} A_{v_i} A_{v_j} = 0] \Pr[A_{v_j} = 0] - (1-c)^2 n \geq \theta n \\
\iff \quad & (1-c)n - \mathbb{E}[\sum_{i=1}^{n} A_{v_i} \mid A_{v_j} = 0]c - (1-c)^2 n \geq \theta n \\
\iff \quad & \mathbb{E}[\sum_{i=1}^{n} A_{v_i} \mid A_{v_j} = 0]c \leq c(1-c)n - \theta n \\
\iff \quad & (1-c)n - \mathbb{E}[\sum_{i=1}^{n} A_{v_i} \mid A_{v_j} = 0]c - (1-c)^2 n \geq \theta n \\
\iff \quad & \sum_{i \neq j} \mathbb{E}[A_{v_i} \mid A_{v_j} = 0] \leq n \left((1-c) - \frac{\theta}{c} \right).
\end{aligned}$$

We conclude that we can find j such that $\sum_{i=1}^{n-1} \mathbb{E}[A_{v_i} \mid A_{v_j} = 0] \leq n \left((1-c) - \frac{\theta}{c} \right)$. Let us assume that in phase 2, the edges of $\partial(u)$ are presented in an order that ensures that v_j comes last.

For convenience, let us immediately relabel the vertices so that the edges of $\partial(u)$ are presented in order $e_1 = (u, v_1), \ldots, e_n = (u, v_n)$ to the OCRS. Notice that by choosing the order of the edges in $\partial(u)$ correctly, we have already established equation (B.5) from the proof of Lemma 4.3. But this equation was the only requirement in the proof of Lemma 4.3 to establish its conclusion that:

$$2c + c \exp(c + \theta/c - 1) \leq 1 + o(1).$$

Note that the θ here has a different meaning than in the proof of Lemma 4.3—here, θ is just a number that satisfies the inequality $\mathrm{Var}[\sum_{i=1}^n A_{v_i}] \geq \theta n^2$. Clearly this inequality is always true for $\theta = 0$, so we conclude that:

$$2c + c \exp(c - 1) \leq 1 + o(1),$$

as was needed.

C Proof of Theorem 5.3

Theorem 5.3 (proven in §C)**.** *There exists a function f satisfying $\lim_{\varepsilon\to 0} f(\varepsilon) = 0$ such that for any 1-regular input $(G(n), \boldsymbol{x}(n))$ with vanishing edge values, and for all $e \in G(n)$, we have:*

$$|\Pr[e \in \mathcal{M} \mid X_e = 1] - \Pr[(u,v) \in \mathcal{M}_{\mathcal{T}}]| \leq f(\varepsilon(n)),$$

where $\mathcal{M}$ is the matching produced by GREEDY-CRS *when run on $G(n)$.*

In order to prove Theorem 5.3, we first restate the following result proven by Nuti and Vondrák (2024) (this is Lemma 1 in (Nuti and Vondrák, 2024)).

Lemma C.1 (Nuti and Vondrák, 2024)**.** *Let T_0 be any finite realization of $\mathcal{T}$. For any edge e of G:*

$$|\Pr[S_e = T_0 \mid X_e = 1] - \Pr[\mathcal{T} = T_0]| \leq 3\varepsilon(n)|T_0|^2.$$

Note that $S_e = T_0$ means that there is an isomorphism from the graphs S_e to T_0 which, if $e = (u_0, v_0)$, sends u_0 to the special vertex $u \in T_0$ and v_0 to the special vertex $v \in T_0$.

We next establish the key proposition needed to prove Theorem 5.3:

Proposition C.2. *Fix any set of finite trees $F = \{T_1, T_2, \ldots, T_m\}$, the following hold:*

1. $|\Pr[S_e \notin F \mid X_e = 1] - \Pr[\mathcal{T} \notin F]| \leq 3\varepsilon \sum |T_i|^2$.
2. $|\Pr[e \in \mathcal{M}, S_e \in F \mid X_e = 1] - \Pr[(u,v) \in \mathcal{M}_{\mathcal{T}}, \mathcal{T} \in F]| \leq 3\varepsilon \sum |T_i|^2$.

Proof of Proposition C.2. For the first result, note that Lemma C.1 implies

$$|\Pr[S_e \in F \mid X_e = 1] - \Pr[\mathcal{T} \in F]| \leq 3\varepsilon \sum |T_i|^2$$

and hence it follows that

$$|\Pr[S_e \notin F \mid X_e = 1] - \Pr[\mathcal{T} \notin F]| \leq 3\varepsilon \sum |T_i|^2.$$

For the second result, notice that since GREEDY-CRS only depends on the active edges of G,

$$\Pr[e \in \mathcal{M} \mid X_e = 1, S_e = T_i] = \Pr[(u,v) \in \mathcal{M}_{\mathcal{T}} \mid \mathcal{T} = T_i].$$

Hence, it follows that

$$\begin{aligned}
& |\Pr[e \in \mathcal{M}, S_e \in F \mid X_e = 1] - \Pr[(u,v) \in \mathcal{M}_{\mathcal{T}}, \mathcal{T} \in F]| \\
= & \left| \sum_{i=1}^{m} \Pr[e \in \mathcal{M} \mid S_e = T_i, X_e = 1] \Pr[S_e = T_i \mid X_e = 1] - \sum_{i=1}^{m} \Pr[(u,v) \in \mathcal{M}_{\mathcal{T}} | \mathcal{T} = T_i] \Pr[\mathcal{T} = T_i] \right| \\
= & \left| \sum_{i=1}^{m} \Pr[(u,v) \in \mathcal{M}_{\mathcal{T}} | \mathcal{T} = T_i] (\Pr[S_e = T_i \mid X_e = 1] - \Pr[\mathcal{T} = T_i]) \right| \\
\leq & \sum_{i=1}^{m} |\Pr[S_e = T_i \mid X_e = 1] - \Pr[\mathcal{T} = T_i]| \\
\leq & 3\varepsilon \sum |T_i|^2.
\end{aligned}$$

□

Proof of Theorem 5.3. First of all, note that the triangle inequality implies

$$\begin{aligned} & |\Pr[e \in \mathcal{M} \mid X_e = 1] - \Pr[(u,v) \in \mathcal{M}_{\mathcal{T}}]| \\ \leq\ & |\Pr[e \in \mathcal{M}, S_e \notin F \mid X_e = 1] - \Pr[(u,v) \in \mathcal{M}_{\mathcal{T}}, \mathcal{T} \notin F]| \\ & + |\Pr[e \in \mathcal{M}, S_e \in F \mid X_e = 1] - \Pr[(u,v) \in \mathcal{M}_{\mathcal{T}}, \mathcal{T} \in F]| \\ \leq\ & \Pr[S_e \notin F \mid X_e = 1] + \Pr[\mathcal{T} \notin F] \\ & + |\Pr[e \in \mathcal{M}, S_e \in F \mid X_e = 1] - \Pr[(u,v) \in \mathcal{M}_{\mathcal{T}}, \mathcal{T} \in F]|, \end{aligned}$$

It follows from Proposition C.2 that:

$$|\Pr[e \in \mathcal{M} \mid X_e = 1] - \Pr[(u,v) \in \mathcal{M}_{\mathcal{T}}]| \leq 2\Pr[\mathcal{T} \notin F] + 6\varepsilon \sum |T_i|^2,$$

and the inequality is true for every F. Now fix a $\delta > 0$, and consider an F with the property that $\Pr[\mathcal{T} \notin F] < \delta/3$. Such an F exists since $\mathcal{T}$ is finite with probability 1. Then, find an ε so small that $6\varepsilon \sum |T_i|^2 < \delta/3$. For such an ε, we must have

$$|\Pr[e \in \mathcal{M} \mid X_e = 1] - \Pr[(u,v) \in \mathcal{M}_{\mathcal{T}}]| \leq 2\Pr[\mathcal{T} \notin F] + 6\varepsilon \sum |T_i|^2 < \delta,$$

and so Theorem 5.3 immediately follows. □

D Deferred Proofs from Section 6

D.1 Proof of Lemma 6.2

Lemma 6.2 (proven in §D.1)**.** *Suppose that $\mathcal{S}$ is a surplus function. Then, $(Q_t)_{t=1}^{n^2}$ as defined in (6.2) is a supermartingale with respect to the filtration $(\mathcal{F}_t)_{t=1}^{n^2}$. Moreover, $\mathbb{E}[|\mathcal{M}_{n^2}|] \leq (1 + o(1))\left(1 - \ln\left(2 - \frac{1}{e}\right)\right) n$.*

Fix $0 \leq t < n^2$ and condition on $\mathcal{F}_t$. Since the FO-CRS is deterministic, the $(t+1)^{th}$ edge $e_{t+1} = (u_{t+1}, v_{t+1})$ is $\mathcal{F}_t$-measurable. Moreover, since e_{t+1} has not yet been processed, the independence of the edge states implies that

$$\Pr[X_{e_{t+1}} = 1 \mid \mathcal{F}_t] = p, \qquad \Pr[X_{e_{t+1}} = 0 \mid \mathcal{F}_t] = 1 - p. \tag{D.1}$$

Suppose first that $e_{t+1} \notin E(R_t)$. By the definition of R_t, the edge e_{t+1} is infeasible and therefore cannot be added to the matching. Since $e_{t+1} \notin E(R_t)$,

$$\mathcal{M}_{t+1} = \mathcal{M}_t, \qquad R_{t+1} = R_t - e_{t+1} = R_t.$$

Consequently, $Q_{t+1} = Q_t$. It remains to consider the case $e_{t+1} \in E(R_t)$. Conditional on $\mathcal{F}_t$, the deterministic FO-CRS either rejects e_{t+1} when it is active or accepts it when it is active. Suppose first that the FO-CRS rejects e_{t+1} when it is active. Then, regardless of the value of $X_{e_{t+1}}$,

$$\mathcal{M}_{t+1} = \mathcal{M}_t, \qquad R_{t+1} = R_t \setminus e_{t+1}.$$

Therefore, by (6.4),

$$Q_{t+1} = |\mathcal{M}_t| + \mathcal{S}(R_t \setminus e_{t+1}) \leq |\mathcal{M}_t| + \mathcal{S}(R_t) = Q_t.$$

Otherwise, the FO-CRS accepts e_{t+1} when it is active. If $X_{e_{t+1}} = 0$, then

$$\mathcal{M}_{t+1} = \mathcal{M}_t, \qquad R_{t+1} = R_t \setminus e_{t+1}.$$

If $X_{e_{t+1}} = 1$, then

$$\mathcal{M}_{t+1} = \mathcal{M}_t \cup \{e_{t+1}\}, \qquad R_{t+1} = R_t - u_{t+1} - v_{t+1}.$$

Thus, after averaging over $X_{e_{t+1}}$ and applying (D.1), we can write $\mathbb{E}[Q_{t+1} \mid \mathcal{F}_t]$ as

$$\begin{aligned}
&\Pr[X_{e_{t+1}} = 0 \mid \mathcal{F}_t](|\mathcal{M}_t| + \mathcal{S}(R_t \setminus e_{t+1})) + \Pr[X_{e_{t+1}} = 1 \mid \mathcal{F}_t](|\mathcal{M}_t| + 1 + \mathcal{S}(R_t - u_{t+1} - v_{t+1})) \\
&= |\mathcal{M}_t| + (1-p)\mathcal{S}(R_t \setminus e_{t+1}) + p\big(1 + \mathcal{S}(R_t - u_{t+1} - v_{t+1})\big) \\
&\le |\mathcal{M}_t| + \mathcal{S}(R_t) = Q_t,
\end{aligned}$$

where the inequality uses (6.5). Thus, no matter the decision of the FO-CRS when processing e_{t+1}, $\mathbb{E}[Q_{t+1} \mid \mathcal{F}_t] \le Q_t$. Hence $(Q_t)_{t=0}^{n^2}$ is a supermartingale with respect to $(\mathcal{F}_t)_{t=0}^{n^2}$.

To complete the proof, recall that $Q_t = |\mathcal{M}_t| + \mathcal{S}(R_t)$ for each $t = 0, \dots, n^2$. Thus, since $(Q_t)_{t=0}^{n^2}$ is a supermartingale, we can take expecations to get that

$$\mathbb{E}[|\mathcal{M}_{n^2}| + \mathcal{S}(R_{n^2})] = \mathbb{E}[Q_{n^2}] \le \mathbb{E}[Q_0] = \mathbb{E}[|\mathcal{M}_0| + \mathcal{S}(R_0)]. \tag{D.2}$$

On the other hand, $\mathcal{S}(R_{n^2}) \ge 0$, $|\mathcal{M}_0| = 0$, and $\mathbb{E}[\mathcal{S}(R_0)] \le (1 + o(1))\left(1 - \ln\left(2 - \frac{1}{\mathrm{e}}\right)\right) n$ by (6.6). Combining these inequalities with (D.2) yields

$$\mathbb{E}[|\mathcal{M}_{n^2}|] \le (1 + o(1))\left(1 - \ln\left(2 - \frac{1}{\mathrm{e}}\right)\right) n,$$

as required.

D.2 Proof of Lemma 6.4

Lemma 6.4 (proven in §D.2)**.** *For every subgraph $H \subseteq G$ and every edge $e = (u, v) \in E(H)$,*

$$\Pr[\mathcal{I}_H] \le (1 - p^2) \Pr[\mathcal{I}_{H \setminus \{e\}}]^q \Pr[\mathcal{I}_{H-u-v}]^p. \tag{6.12}$$

Equivalently,

$$\Pr[\mathcal{I}_H \mid \mathcal{I}_{H-u-v}] \le (1 - p^2) \Pr[\mathcal{I}_{H \setminus \{e\}} \mid \mathcal{I}_{H-u-v}]^q. \tag{6.13}$$

Write $\boldsymbol{B}_{-u-v} := (B_w)_{w \in V(G) \setminus \{u,v\}}$, and call u *available* if $B_w = 0$ for every $w \in N_H(u) \setminus \{v\}$, and define the availability of v analogously. Denote these events by A_u and A_v, respectively, and notice that A_u and A_v are measurable with respect to $\boldsymbol{B}_{-u-v}$, as is the event $\mathcal{I}_{H-u-v}$. In what follows, we condition on $\boldsymbol{B}_{-u-v}$, and assume that we have an instantiation of it for which $\mathcal{I}_{H-u-v}$ occurs. Our goal is to first show that

$$\Pr[\mathcal{I}_H \mid \boldsymbol{B}_{-u-v}] \cdot \mathbf{1}_{\mathcal{I}_{H-u-v}} \le (1 - p^2) \Pr[\mathcal{I}_{H \setminus \{e\}} \mid \boldsymbol{B}_{-u-v}]^q \cdot \mathbf{1}_{\mathcal{I}_{H-u-v}}. \tag{D.3}$$

We separate (D.3) into cases, dependent on which combination of A_u and A_v occur. Note that conditional on $\boldsymbol{B}_{-u-v}$, B_u and B_v remain independent $\mathrm{Ber}(p)$ random variables.

Suppose first that A_u, A_v and $\mathcal{I}_{H-u-v}$ all occur. In this case, for $\mathcal{I}_{H \setminus \{e\}}$ to occur, both B_u and B_v can take any value. On the other hand, for $\mathcal{I}_H$ to occur, the outcome $B_u = B_v = 1$ is forbidden. Consequently,

$$\Pr[\mathcal{I}_H \mid \boldsymbol{B}_{-u-v}] \cdot \mathbf{1}_{A_u \cap A_v} \mathbf{1}_{\mathcal{I}_{H-u-v}} = 1 - p^2, \qquad \Pr[\mathcal{I}_{H \setminus \{e\}} \mid \boldsymbol{B}_{-u-v}] \cdot \mathbf{1}_{A_u \cap A_v} \mathbf{1}_{\mathcal{I}_{H-u-v}} = 1,$$

and so (D.3) follows immediately.

Now suppose that $\mathcal{I}_{H-u-v}$ occurs, and at least one of u and v is unavailable. In order for $\mathcal{I}_{H\setminus\{e\}}$ to occur, if $\neg A_u \cap \neg A_v$ holds, then we must have $B_u = B_v = 0$. Similarly, if $\neg A_u \cap A_v$ (respectively, $A_u \cap \neg A_v$) holds, then we must have $B_u = 0$ (respectively, $B_v = 0$). The exact same cases hold for $\mathcal{I}_H$. Thus,

$$\Pr[\mathcal{I}_H \mid \boldsymbol{B}_{-u-v}] \cdot \mathbf{1}_{\neg A_u \cup \neg A_v} \mathbf{1}_{\mathcal{I}_{H-u-v}} = \Pr[\mathcal{I}_{H\setminus\{e\}} \mid \boldsymbol{B}_{-u-v}] \cdot \mathbf{1}_{\neg A_u \cup \neg A_v} \mathbf{1}_{\mathcal{I}_{H-u-v}} \in \{q^2, q\}. \tag{D.4}$$

Weighted AM–GM, with weights p and q, gives $q^p = q^p 1^q \le pq + q \cdot 1 = 1 - p^2$, and so for every $x \in [0, q]$,

$$x = x^q x^p \le q^p x^q \le (1 - p^2) x^q. \tag{D.5}$$

By taking $x \in \{q, q^2\}$ we can apply (D.4) with (D.5), to get (D.3).

To complete the proof, we take conditional expectations with respect to $\mathcal{I}_{H-u-v}$ over (D.3) to get that

$$\begin{aligned}
\Pr[\mathcal{I}_H \mid \mathcal{I}_{H-u-v}] &= \mathbb{E}\left[\Pr[\mathcal{I}_H \mid \boldsymbol{B}_{-u-v}] \mid \mathcal{I}_{H-u-v}\right] \\
&\le (1-p^2)\mathbb{E}\left[\Pr[\mathcal{I}_{H\setminus\{e\}} \mid \boldsymbol{B}_{-u-v}]^q \,\middle|\, \mathcal{I}_{H-u-v}\right] \\
&\le (1-p^2)\mathbb{E}\left[\Pr[\mathcal{I}_{H\setminus\{e\}} \mid \boldsymbol{B}_{-u-v}] \,\middle|\, \mathcal{I}_{H-u-v}\right]^q \\
&= (1-p^2)\Pr[\mathcal{I}_{H\setminus\{e\}} \mid \mathcal{I}_{H-u-v}]^q.
\end{aligned}$$

Here the first equality uses that $\mathcal{I}_{H-u-v}$ is measurable with respect to $\boldsymbol{B}_{-u-v}$. The second inequality is (D.3) with the conditional expectation applied. The third inequality uses Jensen's inequality, as $x \mapsto x^q$ is concave. The final equality once again uses that $\mathcal{I}_{H-u-v}$ is measurable with respect to $\boldsymbol{B}_{-u-v}$. This proves (6.13).

Since $\mathcal{I}_H$ and $\mathcal{I}_{H\setminus\{e\}}$ both imply $\mathcal{I}_{H-u-v}$, we have

$$\Pr[\mathcal{I}_H \mid \mathcal{I}_{H-u-v}] = \frac{\Pr[\mathcal{I}_H]}{\Pr[\mathcal{I}_{H-u-v}]}, \qquad \Pr[\mathcal{I}_{H\setminus\{e\}} \mid \mathcal{I}_{H-u-v}] = \frac{\Pr[\mathcal{I}_{H\setminus\{e\}}]}{\Pr[\mathcal{I}_{H-u-v}]}.$$

Substituting these identities into (6.13), multiplying by $\Pr[\mathcal{I}_{H-u-v}]$, and using that $1-q = p$ proves (6.12).

E Concentration Property of OCRSs

Proposition E.1. *Fix an input $(G, \boldsymbol{x})$ and an arrival order of the edges E of G. Suppose that, for each arriving edge e, an OCRS independently draws*

$$A_e \sim \mathrm{Ber}(a_e),$$

where $a_e \in [0,1]$ is deterministic once $(G, \boldsymbol{x})$ and the arrival order are fixed. The OCRS adds e to its current matching if and only if $X_e = A_e = 1$ and both endpoints of e were previously unmatched.

Consider a sequence of inputs $(G(n), \boldsymbol{x}(n))$ such that if $G(n) = (V(n), E(n))$, then as $n \to \infty$,

$$s_n := \sum_{e \in E(n)} x_e(n) \longrightarrow \infty.$$

If the resulting OCRS is c_n-selectable on $(G(n), \boldsymbol{x}(n))$, where $\liminf_{n\to\infty} c_n > 0$, then it is concentrated on this sequence of inputs. That is, if M_n denotes the matching it outputs on G_n, then w.h.p.,

$$\big||M_n| - \mathbb{E}[|M_n|]\big| = o\big(\mathbb{E}[|M_n|]\big).$$

Proof. Fix $n \in \mathbb{N}$. For convenience, we suppress the dependence on n from the notation. Since the arrival order is fixed, we may equivalently draw all the random variables $(A_e)_{e\in E}$ before the execution begins. Define

$$Y_e := A_e X_e.$$

The random variables $(Y_e)_{e\in E}$ are mutually independent, with

$$q_e := \Pr[Y_e = 1] = a_e x_e \le x_e.$$

The matching M is precisely the matching obtained by greedily processing, in the given arrival order, those edges e for which $Y_e = 1$.

We first observe that changing a single coordinate of $\boldsymbol{Y} = (Y_e)_{e\in E}$ changes the cardinality of the resulting matching by at most one. To see this, consider two executions whose candidate vectors differ only on one edge e. The executions are identical before e is processed. If the differing value of Y_e does not cause e to be added in exactly one execution, the two executions remain identical. Otherwise, immediately after processing e, the symmetric difference of the two matchings consists only of e and is therefore an alternating path.

This property is preserved as the remaining edges are processed. Indeed, suppose that a later candidate edge f is added in exactly one of the two executions. Then f is feasible in one execution and infeasible in the other. Consequently, one of its endpoints is unmatched in the first matching and matched in the second matching, and hence is an endpoint of the current alternating path. Adding f therefore extends the alternating path or closes it into an alternating cycle. Thus, throughout the coupled executions, the symmetric difference is empty, an alternating path, or an alternating cycle. The cardinalities of two matchings whose symmetric difference has this form differ by at most one. Hence, writing $M(\boldsymbol{y})$ for the greedy matching obtained from a deterministic candidate vector $\boldsymbol{y}$, we have

$$\big||M(\boldsymbol{y})| - |M(\boldsymbol{y}')|\big| \le 1 \tag{E.1}$$

whenever $\boldsymbol{y}$ and $\boldsymbol{y}'$ differ in a single coordinate.

For each $e \in E$, let $\boldsymbol{Y}^{(e)}$ be obtained from $\boldsymbol{Y}$ by replacing Y_e with an independent copy Y_e'. Applying the Efron–Stein inequality and then (E.1), we obtain

$$\begin{aligned}
\mathrm{Var}(|M|) &\le \frac{1}{2}\sum_{e\in E} \mathbb{E}\left[\Big(|M(\boldsymbol{Y})| - |M(\boldsymbol{Y}^{(e)})|\Big)^2\right] \\
&\le \frac{1}{2}\sum_{e\in E} \Pr[Y_e \neq Y_e'] \\
&= \sum_{e\in E} q_e(1-q_e) \\
&\le \sum_{e\in E} x_e = s_n.
\end{aligned}$$

On the other hand, c_n-selectability implies

$$\mathbb{E}[|M|] = \sum_{e\in E} \Pr[e \in M] = \sum_{e\in E} x_e \Pr[e \in M \mid X_e = 1] \ge c_n s_n.$$

Let $\delta_n := s_n^{-1/4}$. Chebyshev's inequality now gives

$$\begin{aligned}
\Pr\left[\big||M| - \mathbb{E}[|M|]\big| > \delta_n \mathbb{E}[|M|]\right] &\le \frac{\mathrm{Var}(|M|)}{\delta_n^2 \mathbb{E}[|M|]^2} \\
&\le \frac{s_n}{\delta_n^2 c_n^2 s_n^2} = \frac{1}{c_n^2\sqrt{s_n}} = o(1).
\end{aligned}$$

Since $\delta_n \to 0$, this proves w.h.p. $\big||M| - \mathbb{E}[|M|]\big| = o\big(\mathbb{E}[|M|]\big)$. □

For the tree input $(G, \boldsymbol{x})$ from Subsection 4.2 with edges E, we have

$$\sum_{e \in E} x_e = n(n-1)\frac{1}{n} + n\frac{1}{n} = n.$$

Consequently, every OCRS which satisfies the properties of Proposition E.1 is concentrated on this sequence of inputs. In particular, Algorithm 1 belongs to this class, since on $(G, \boldsymbol{x})$, it independently draws

$$A_e \sim \mathrm{Ber}\left(\frac{\alpha}{(1-\alpha x_u(e))(1-\alpha x_v(e))}\right)$$

and adds e precisely when $X_e = A_e = 1$ and its endpoints are unmatched. This is also true of the algorithm of Ezra et al. (2022); MacRury et al. (2025), despite the parameters of its relevant A_e bits not having a simple closed form.

We note that the Gibbs distribution based OCRS for matchings introduced by Aminian et al. (2026) also satisfies the concentration property, using a different argument. Indeed, the distribution μ of matchings selected by their algorithm is of the form:

$$\mu(M) \propto \prod_{e \in M} w_e,$$

where $(w_e)_{e \in E}$ are positive real values, and $\propto$ indicates equality up to a normalization constant. By the Heilmann–Lieb Theorem (see Heilmann and Lieb (1972)), the cardinality generating polynomial

$$\sum_{M} z^{|M|} \prod_{e \in M} w_e$$

has only real negative roots. It follows that $|M|$ is distributed as a sum of independent Bernoulli random variables, and hence

$$\mathrm{Var}(|M|) \leq \mathbb{E}|M|.$$

Therefore, as $\mathbb{E}|M| \to \infty$, we have w.h.p. $\big||M| - \mathbb{E}|M|\big| = o\left(\mathbb{E}|M|\right)$.